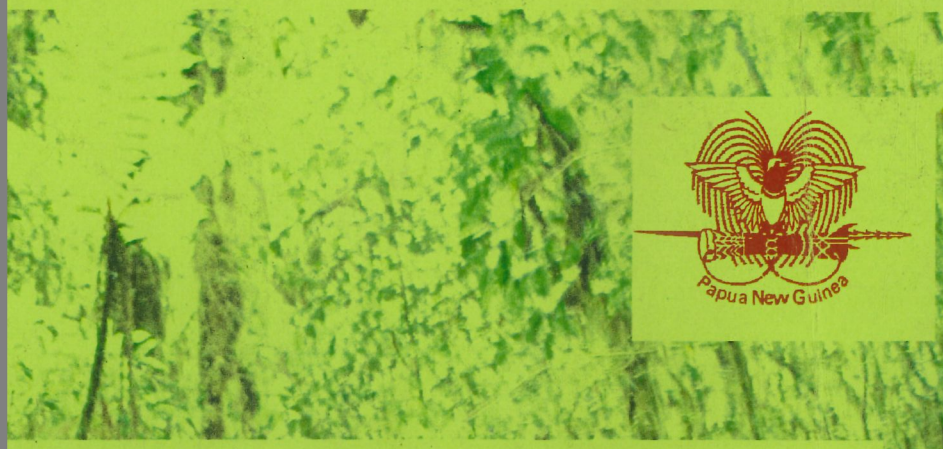
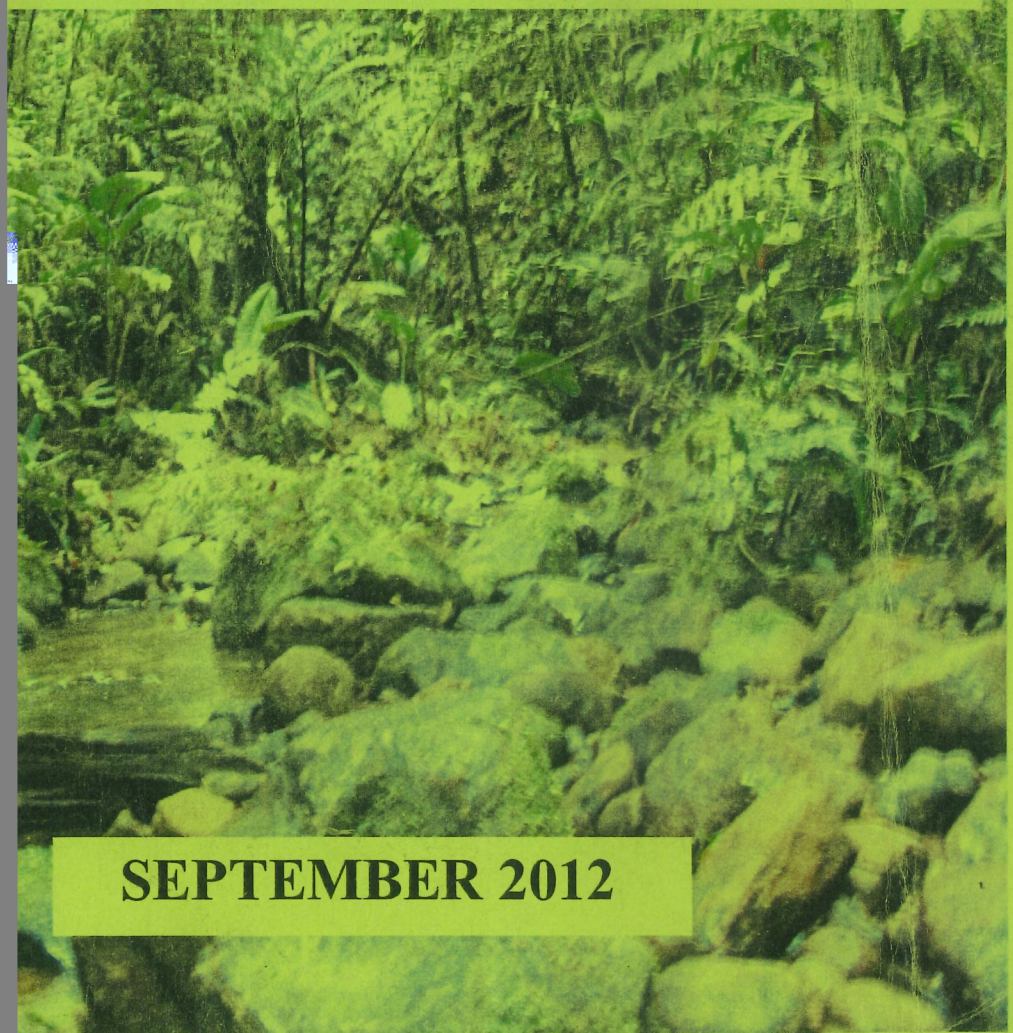
GUINEA FOREST AUTHORITY



ONAL FOREST PLAN

PREPARED BY
V GUINEA FOREST AUTHORITY



SEPTEMBER 2012

INTRODUCTION

A National Forest Plan (NFP) is a legislative requirement under the Forest Act 1991 (as amended) Section 47, Subsection 1 that reads, *"The Authority shall cause to be drawn up a NFP to provide a detailed statement of how the National and Provincial Governments intend to manage and utilize the country's forest resources"*

Furthermore Section 47, Subsections (2) to (5) states that;

(2) The National Forest Plan shall

(a) Be consistent with the National Forest Policy and relevant government policies; and

(b) Be based on a certified National Forest inventory which shall include particulars as prescribed; and

(c) Consist of;

i) National Forest Development Guidelines prepared by the Minister in consultation with the Board and endorsed by the National Executive Council; and

ii) National Forest Development Programme; and

iii) A statement prepared annually by the Board of annual cut volumes, being the amount of allowable cut for each province for the next succeeding year which will ensure that the areas of forest resource set out in the Provincial Forest Plan, for present or future production, are harvested on a sustainable yield basis.

iv) Provincial Forest Plans

(3) For the purpose of Subsection (2) (c) (iii), 'allowable cut' means the amount of timber which can be cut annually.

(4) The NFP shall be reviewed every five years from the date of coming into operation of the Forestry (Amendment) Act, 2005 and shall be submitted to the National Executive Council (NEC) for its approval.

(5) Where the NFP is not approved by the NEC, it shall be referred to the Authority for review and recommendation for approval to the NEC which shall grant its approval.

Since the release of the first NFP in November 1996, a lot has happened in the forestry sector and this review is to reflect on the past and plan for the future based on the changes that have taken place. Much of the accessible forests areas have been logged out or cleared for agriculture projects and other land uses. Only minimal accessible forest areas are remaining while much of it is in the hinterland where there is limited access. The issue of climate change has taken centre stage in recent years where forests are considered to be the cause and cure and PNG is playing an active role in mitigating the impact of climate change.

Many timber project areas have merchantable or commercial trees logged out while only a handful of new projects have come into operation. Others are yet to be inventoried and acquired for timber harvesting. Most if not all older timber projects acquired under the provision of the old Forest Act Chapter 216, resulting in a Timber Rights Purchase (TRP) or the Private Dealings Act Chapter 217, resulting in a Local Forest Area (LFA) prior to the concept of Forest Management Agreement (FMA) under the current Forest Act 1991 were mostly not sustainable forestry operations.

The overall forestry sector in PNG has received much attention and commentary by national and international NGO's, Government's, private organizations, individuals and other stakeholders. Additionally the pace of forest resource exploitation and or development has increased dramatically resulting in the decline in accessible forest resources areas. The global economic crisis in 1997 drastically affected the prices of logs as well as other goods and services thereby resulting in the closure of a number of logging operations in the country. The closure was also attributed in part to the depletion of forest resources as those earlier projects were not meant to be sustainable projects but vehicles for delivery of goods and services to the rural people.

Other global issues such as climate change, biodiversity conservation, forest certification etc, have become major issues facing the international communities at large and impact also on the forestry sector of PNG. These are noted toward the end of this document and shall be given due consideration in the next review or sooner.

The NFP shall now be reviewed every 5 years as per the Forestry Act 1991 (as amended).

1.0 FOREST POLICY OBJECTIVES

The Forest Policy is guided and based on the fourth goal of the constitution which states: *"For Papua New Guinean's natural resources and the environment to be conserved and used for the collective benefit of us all, and to be replenished for the benefit of future generations".*

Subsequently two main and three supportive objectives were formulated aimed at the meeting the national goal.

The 1991 National Forest Policy's main objectives are;

i. Management and protection of the nation's forest resource as a renewable natural assets.

ii. Utilization of the nation's forest resources to achieve economic growth, employment creation, greater Papua New Guinean participation in industry and increased viable onshore processing.

Supportive objectives are:

i. The collection of information and advancement of knowledge relating to the utilization and maintenance of Papua New Guinea's forest resources through forest research.

ii. Increased acquisition and dissemination of skills, knowledge and information in forestry through education and training.

iii. Effective strategies, including administrative and legal machinery, to manage the forest resources, and incorporating national, provincial and local interests.

The above main policy objectives and sub-objectives are the basis for the National Forest Plan (NFP) that is aimed at delivering the stated objectives through programmes, projects and activities.

3.0 NATIONAL FOREST DEVELOPMENT GUIDELINES

The 2009 National Forest Development Guidelines (NFDG) also provides the basis for the sound management of the nation's forest resources taking into account the success, failures and the emerging issues. It provides the road map of how the Government wishes to see the forestry sector manage in the medium term. The 2009 NFDG draws strength from the National Forest Policy in realizing its objectives of sustainable management of the forest resources. It emphases the importance of a more intense forest management prescription with establishing Permanent Forest Estate (PFE) through the natural forest management and plantation forest development, particularly in strategic locations of the country to enable social, economical and environmental sustainability.

The main thrust of the Government's initiative has and continues to be towards achieving economic recovery and nation building through the development of the nation's natural resources including the forest resources. In doing so the Government bears in mind the need for environmental preservation, social and cultural values of the forests in PNG. This can be achieved through among other means, sustainable forest management and optimizing financial benefits from any forest based projects.

The 2009 National Forest Development Guidelines is aimed to:

i. Improve the capacity of the public administrative unit, the PNG Forest Authority, to fulfill its role under the National Forest Policy and as contained in the objectives of the Forestry Act.

ii. Promote the input of the resource owners, local and provincial interests in co-ordination with reliable investors to engage in and better develop secure resource access, protection and sustainability of based industries and a designated Permanent Forest Estate for Papua New Guinea.

iii. Develop a focus in wood and non-wood products and in relation to future carbon based schemes for adaptation and mitigation action to combat climate change.

iv. Assist to develop diversified markets and encourage the development of national enterprises in the forestry sector.

v. While maintaining the existing strong investment base, promote general diversification in forest sector private investment, based on merit in the demonstration of, or capacity for, performance to compliance criteria.

vi. Maintain a healthy and secured resource based environment conducive to attracting and retaining a private enterprise investment base which is efficient and capable and which promotes good corporate governance and citizenship.

vii. Establish a uniform and capable management structure and administration of the forest sector in Papua New Guinea while acknowledging the diversity of forest dynamics, and the existing variation in the objectives of local economic, cultural and social influences.

viii. Develop strong corporate public sector leadership and management within the PNG Forest Authority, which is fashioned as the co-ordinator of all forest based activity in the national interest.

It is intended to provide for the orderly development of the forestry sector to confirm to Government's policy directives of the day and is subject to change. The National Forestry Development Guidelines (NFDG) forms one of the bases of the NFP

Apart from the NFDG, the National Government's Development Strategies and Vision 2050 will also provide some form of guidelines for the Forest Authority to sustainably manage our forest resources.

While it is not attached herewith, the NFDG can be a separate document being available from the PNGFA.

4.0 PROVINCIAL FOREST PLANS

It is a requirement under Section 49 of the Forestry Act 1991 (as amended) that each province shall as soon as practical have in a place a Provincial Forest Plan (PFP) that will state how the province wishes to manage its forest resources.

The National Forest Plan (NFP) shall then be a summation of all the Provincial Forest Plans (PFPs). For practical reasons the NFP includes only those areas that are planned for forest resource development in the provinces. Besides forest related activities, protected and other significant areas for conservation are also captured in the plans, therefore concerned agencies should make reference to these plans prior to implementing their programs. It would be useful that these agencies and other stakeholders make reference to the Provincial Forest Plans.

Unless an area of potential is listed in the PFP there shall be no inventory or other activity leading to an FMA and project development undertaken over the area. Additionally it should conform to the NFDG.

Over the medium term PNGFA shall ensure that every province continues to have in place a current PFP that will reflect the aspirations of the province on how they wish to have their forest resources managed and developed. These plans will be subject to review every 5 years and then incorporated into the review of the NFP.

Under the 2005 amendment to the Forest Act, all PFP's are required to be reviewed every fifth year as in the past when these were expected to be reviewed every 3 years. Furthermore, under the same amendment the NFP is now a summation of the 19 provincial forest plans.

For more details on the Provincial Development Programmes, reference should be made to the respective Provincial Forest Plans.

5.0 FOREST AND LAND CLASSIFICATION

5.1 FOREST CLASSIFICATION

The Forest Policy stipulates that the forests of PNG be classified into the following broad categories that reflect more on the end use of the forest rather than the biological classification of forest types. These categories are;

- Production Forest – identified as timber production areas in the long term.
- Protection Forests – by virtue of their location, topographic constraints, and ecological, cultural, or environmental considerations.
- Reserve Forests – not yet otherwise classified, but upon which a decision will be reached later.
- Salvage Forest – forest to be cleared for other land use.
- Land suitable for reforestation.

These areas are, and shall continue to be identified by the Provincial Govemments with the assistance of the PNGFA. When designating these areas, consideration is given to the multiple use of the forest including but not limited to environmental consideration, recreation, provision of subsistence and cultural requirements and the supply of timber and other forest products.

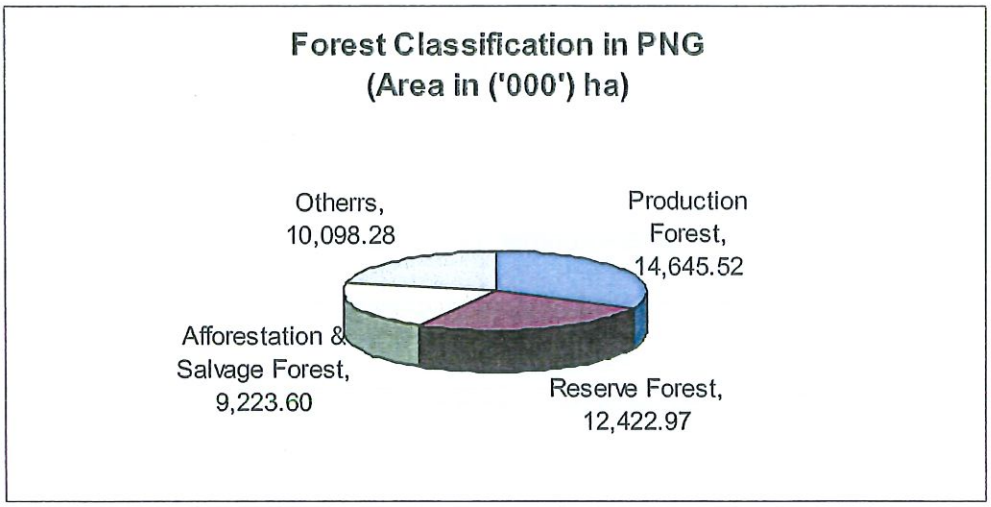
The forest classification is shown below in **Table 1** and **Figure 1**. Other information is provided in **Appendix 2** and also in the **Forest Cover and Land use Map of PNG.**

Table 1: Forest Classification

Forest Classification (Function)	Area ('000ha)	Remarks
Production	14,645.52	Identified as timber production areas in the long term. An estimated total of 5.5 million ha is available for development under concessions while 9.1 million ha is considered as future production forest.
Reserve Forest and (Protection Forests)	12,442.97	While decisions have yet to be made on their functions, a large proportion would be for protection although only 1.796 million hectares have been either been officially as conservation areas and or being proposed. Other forested areas are classified as protection forest by virtue of their location, topographic constraints, and ecological, cultural or Environmental considerations.
Afforestation & Salvage Forests	9,223.60	Grassland areas and forests cleared for other uses.
Others	10,098.29	Areas taken up by other land-use including urban settlements, agriculture and so forth
TOTAL	46,410.38	

Note: Due to insufficient information, areas for afforestation and salvage forest cannot be classified separately.

Figure 1: Forest Classification of PNG



Source: FIMS DATABASE

5.2 LAND CLASSIFICATION

Land has also been classified using the current PNG Forest Authority's Forest Inventory Mapping Systems (FIMS) into categories primarily to differentiate between those areas that have to be excluded from logging operations due to the nature of their location. Factors such as altitude, topography, geology, and level of inundation form the basis for this classification.

More information on the land and type of constraints is shown in **Appendix 2.**

6.0 NATIONAL FOREST INVENTORY

The National Forest Board of PNGFA in 2005 endorsed the Forest Inventory Mapping Systems (FIMS) as the official National Forest Inventory (NFI) of PNG. This internal PNGFA data base commonly referred to as FIMS was developed in 1997/98 and meets the requirements as prescribed by both the Forest Policy 1991 and subsequent Forestry Regulation 1998. It generally contains all the prescriptions of an NFI except for the capture of minor forest products and a growth and yield model. The latter called the 'PINFORM' model is yet to be tested by FRI before it can be use for growth and yield prediction.

The appendices and tables in this plan have been sourced from the FIMS database, for example **Appendix 1** relating to the "Forest Resource of PNG" and **Table 1** referring to the "Forest Classification of PNG".

The FIMS database will be subject to changes as land use and other relevant information is updated from time to time. Besides forest related activities, protected and other significant

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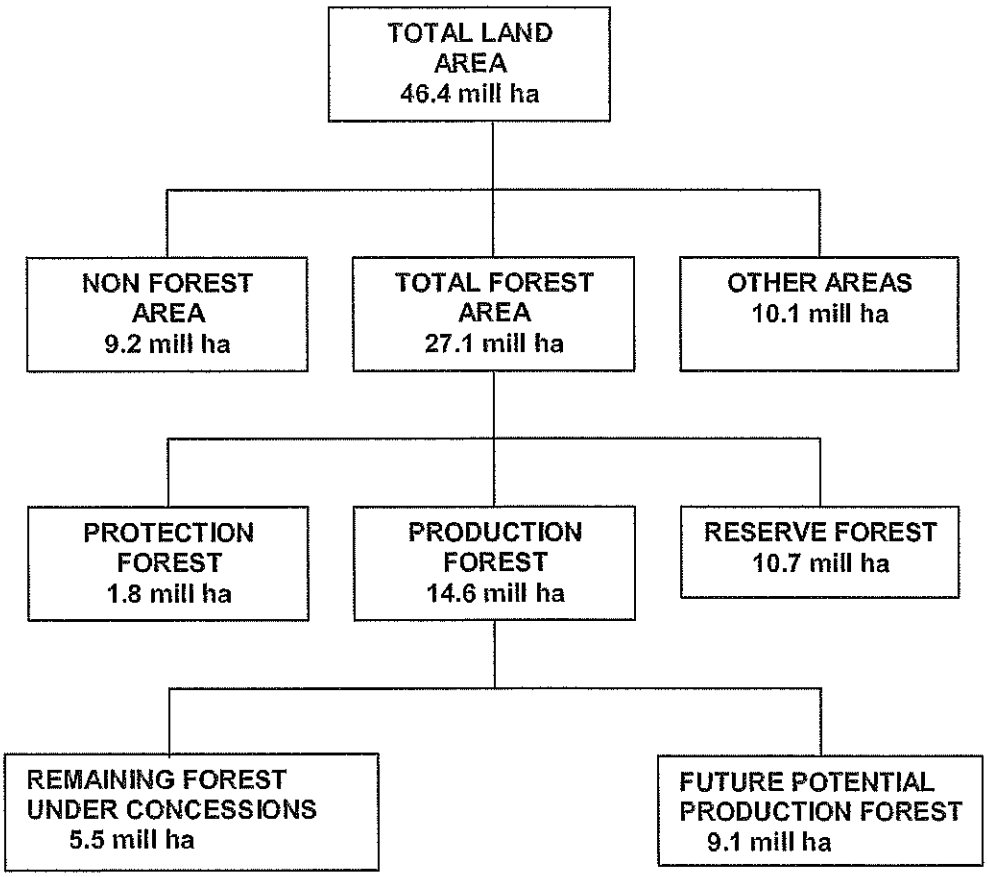
development increases. Logging, shifting cultivation, commercial agriculture, urban development, infrastructure development, mining, bush fires and various natural disasters were identified to be the main drivers of deforestation and forest degradations[1].

Small scale sawmilling operations using portable sawmills including large scale FCA operations also contributes to the loss of forest.

Figure 2: Forest Resources of PNG



Various reports also gave different percentages of the deforestation rates per annum; 0.5 percent (FRA, 2005)[2] and 1.41 percent (Shearman et al. 2008). Forest accessible to logging is estimated by Shearman to be degraded or cleared at a faster rate of 2.6 percent (362,400 ha/annum) hence if the trend continues, it is estimated that by 2021, 83% of the accessible forest areas will be degraded. Due to these differences, there is a need for PNG's forest area to be verified possibly using remote sensing technology in order for appropriate actions to be taken.

[1] Sherman P., et al. 2008.The State of the Forests of Papua New Guinea, Port Moresby, University Of Papua New Guinea

[2] Global Forest Resource Assessment 2005,FAO

3 there were approximately 10.622 million hectares of forest under some ; Timber Rights Purchase (TRP) area, Local Forest Area (LFA) and Forest ent (FMA). Currently as the older timber permits continue to expire, the under timber permits is estimated at 8.5 million hectares. This trend is ue as more concessions under TRP including LFA expires and no new e acquired

the whole seems to have a lot of forest, much of these forests are in sible/inoperable areas including water catchment, swamp forests and as the central ranges and the number of mountain chains that emanate e. This will make them unattractive to timber development but will have an cting areas of water catchment, wildlife habitats and so forth.

nt that sound planning be undertaken for the remaining forest resources to objectives are fulfilled. It is also important that a comprehensive survey is ine the changes and the size of the overall forest area to allow for future

ange is shown in **Appendix 1**.

CESSIONS IN PNG

PNG are owned by the customary land groups and in order for any form of to take place, there has to be an agreement in place between the landowners. In the past these agreements were executed under the old 16 resulting in a "Timber Rights Purchase" or under the Private Dealings al Forest Area". Many of the expired concessions were the result of the tion through either one of these legal mechanisms.

t of the Forestry Act 1991 (as amended), the system through which land tained is through the Forest Management Agreement (FMA) pursuant to and 58 of the Forestry Act 1991 (as amended). FMA is a contract ners and the PNGFA through incorporated land group (ILG) process in s agree to their forest being managed sustainably. Under this agreement quired for a period of 50 years.

gal means to harvest timber from customary land and that is through the er Authority" (TA) as provided for under the Forestry Act 1991 (as tly for short term period (annually) and restricts the harvestable volume to omestic processing. The other form of log harvest is through a "Forest FCA) purposely created for agriculture and road line clearance where the exported.

, a total of 11.0 million hectares of land were under some form of timber those that have expired. Currently 8.5 million hectares is under some Of this total, 5.0 million hectares is under Forest Management Agreement e core of timber concessions that would be developed on a sustainable low).

Other areas operating under Timber Authority cannot be verified due to the type and nature of the operations however areas acquired for major agriculture projects under FCA is estimated at 0.950 million hectares.

Considering that there is technically no area designated as Permanent Forest Estate (PFE) besides the plantations being grown on state land, FMAs shall continue to serve such a role upon which the Annual Allowable Cut (AAC) shall be determine.

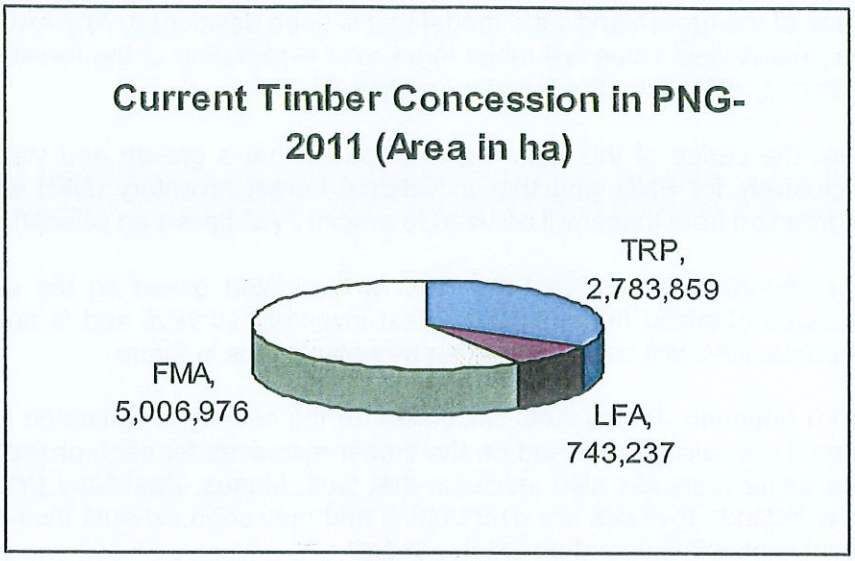
TABLE 2: SUMMARY OF TIMBER CONCESSIONS IN PNG-2011

PROVINCE	TRP (ha)		LFA (ha)		FMA (ha)	TOTAL(ha)
	Current	Expired	Current	Expired		
Western	722,795	24,289	0	0	498,205	1,245,289
Gulf	373,303	142,192	0	0	1,769,834	2,285,329
Central	200,705	121,388	45,000	0	173,190	540,283
Milne Bay	31,340	200,809	0	0	82,380	314,529
Oro	29,620	96,233	39,930	0	221,000	386,783
S.Highlands	0	39,241	0	0	98,750	137,991
Eastern Highlands	0	4,589	0	0	0	4,589
Chimbu	0	0	0	0	0	Nil
W.Highlands	0	84,438	0	0	0	84,438
Jiwaka	0	0	0	0	0	
Enga	0	43,568	0	0	0	43,568
Hela	0	0	0	0	0	
West Sepik	0	150,103	48,500	0	901,974	1,100,577
East Sepik	0	29,470	79,978	0	521,500	630,948
Madang	131,480	159,392	0	0	253,500	544,372
Morobe	67,239	242,952	0	0	131,760	441,951
West New Britain	669,355	285,867	491,898	0	354,883	1,802,003
East New Britain	189,520	391,830	0	0	0	581,350
New Ireland	289,115	98,411	0	243,145	0	630,701
ARB	46,720	78,246	0	0	0	124,966
Manus	32,667	2,496	37,931	0	0	73,094
TOTAL	**2,783,859**	**2,195,544**	**743,237**	**243,145**	**5,006,976**	**10,972,761**

Source: Forest Resource Acquisition General Information, December, 2011

Note: Jiwaka and Hela are new provinces.

Figure 3: Current Timber Concessions in PNG



Detail information on the concession areas and their status can be obtained from the **Forest Resource Acquisition General Information Update, December 2011,** available at the Policy and Planning Directorate. Additional information is given in **Table 2** and **Appendix 4.**

9.0 STATEMENT OF ANNUAL ALLOWABLE CUT

The preparation of a Statement of Annual Allowable Cut (AAC) as a component of the NFP is a requirement under Section 47 (2)I(iii) of the Forestry Act 1991 (as amended) and reads as follows *"A statement, prepared annually by the Board, of allowable cut volumes, being the amount of allowable cut for each province for the next succeeding year which will ensure that the areas of forest resource set out in the Provincial Forest Plan, for present or future production, are harvested on a sustainable yield basis"*

Under the National Forest Policy 1991, it is also clearly stipulated that;

The National Forest Board shall determine the annual allowable cut for each province at a level which will ensure that the areas of forests earmarked in the Provincial Forest Plan for the present or future production are harvested on a sustainable basis. The allowable cut volumes will be revised as necessary to take account of refinements in the National Forestry Inventory's database.

Where existing or proposed harvest volumes in a province exceed the AAC volume, the permitted harvest volumes for the production areas in the province will be reduced accordingly. The new permitted volumes per area will be set by the National Forest Board acting on the advice of the Provincial Forest Management Committees (PFMC).

Current operations and resource areas may be consolidated where this is practical in order to achieve economies of scale in single reforestation, harvesting or processing operations under revised harvest volumes.

The AAC shall be revised annually based on new information coming to hand and the final out come of the growth and yield model that is been developed. Any AAC projections that are above the default yield value will result in an over exploitation of the forest resource thereby making it extremely difficult for the forest to recuperate.

Over the period of this plan it is anticipated that a growth and yield model will be developed exclusively for PNG and that a National Forest Inventory (NFI) will be updated so that the information from these will be used to predict AAC based on scientific information.

It is important to note that the AAC is calculated based on the estimated volume of timber resource obtained from PNGFA forest inventory surveys and is based on natural forest only. Separate AAC will be calculated for tree plantations in future.

From Appendix 4, total AAC calculated for the country is estimated to be 10.5 million m^3. Actual annual cut calculated based on the timber resources for each project for 2012 is 8.9 million m^3. The same appendix also indicates that Gulf, Manus, East New Britain, West New Britain and New Ireland provinces are over cutting and may soon exhaust their resources if the same level of cut is maintain over the next five years.

The projected AAC for the provinces for a period of five years from 2013 to 2017 is shown in **Appendix 5a.** Additional information is provided in **Appendix 5b.** Estimated AAC for the Future Potential Forest Development Areas are not included in the projection but is shown in **Appendix 5c.**

10.0 FOREST RESOURCE DEVELOPMENT PROGRAMME

10.1 NATIONAL FOREST DEVELOPMENT PROGRAMME

The Forest Development Programme (FDP) is based on the 19 Provincial Forest Plans. It deals only with the areas that have been identified in the plans as areas having potential for forest development. It is based on these areas that a National Forest Development Programme (NFDP) is prepared. The rationale is that no activities leading to a FMA and ultimately forestry project development shall be entertained unless it is listed in the Provincial and the National Forest Plan.

All areas referred to as Potential Forest Development Areas (PFDAs) are those forest areas that have been identified by the provinces and listed in their respective PFPs for future development.

PFDAs will not automatically be developed as forestry projects as a lot more work would have to be done initially to ascertain the forest resource to justify a sustainable forest operation and also if there are environmental constraints that may prohibit logging operations including resource owners consent.It is these PFDAs that form the basis for this National Forest Development Programme 2013-2017.

The Potential Forest Development Areas are listed in **Appendix 5C** and also under the National Forest development Programme in **Appendix 6.** Detail information on these areas can be obtained from the respective PFPs.

The National Forest Development Programme comprises the following activities;

10.11 Forest Inventory

A programme for inventory over potential forest areas including its associated budget estimates as identified in the PFPs are listed in **appendix 6**.The implementation of these programme shall be dependent on funding that is made available to the PNGFA in either recurrent or Public Investment Program (PIP) budget or through donor funding. It can also be funded by the respective provincial governments and or through elected members of parliament.

The overall costing for the National Forest Inventory is estimated at **5.7 million kina.**

10.12 Forest Resource Acquisition

Forest Acquisition shall continue to be in the form of a Forest Management Agreement (FMA) in compliance with the Forestry Act 1991 (as amended). It shall be undertaken over areas that have been identified and listed in the PFP's and subsequently the NFP. The ability of the PNGFA to implement this programme shall also depend on its annual budgets.

Landowner awareness is an important and integral part of forest acquisition programme as no FMA can be executed unless there is consent by the landowners. Following awareness in the proposed PFDAs, land groups are formed in accordance with the Land Group Incorporation Act (Amendment) 2009 after which an FMA can be executed

Recognizing the fact that there is limited accessible forest areas remaining, the acquisition programme shall concentrate on priority areas. Many of the PFDAs listed in the programme are small in terms of their estimated AACs therefore are given lower priority however resource owners may apply for Timber Authority (TA) for harvesting and downstream processing.

The cost of conducting the ILGs and finalizing the FMAs is estimated at **5.8 million kina.**

10.13 Forest Allocations

The activities leading up to the allocation of forestry projects incurs huge costs to the PNGFA and shall be implemented according to the National Forest Development Programme and guided by the National Forest Development Guidelines. The PNGFA shall concentrate its resources in ensuring that potential forest areas are inventoried, acquired and allocated, subject of course to land owner consents.

Appendix 5 indicates the estimated costs for conducting Development Options Studies (DOS) including the costs involved in the allocation process for each of the potential projects.

10.2 FOREST RESOURCE REPLACEMENT

Under the 1991 National Forest Policy, the following reference is made to reforestation;

(a) Reforestation will be promoted as a means to maintain Permanent Forest Estate to supply wood to existing and new forest industries. Programmes for plantation development will be guided by economic criteria and feasibility studies to assess the commercial viability of processing plantation material for a variety of end uses.

(b) A programme of non-forested and degraded land shall be pursued and developed.

(b) Woodlot establishment, agro-forestry and tree planting programmes shall be contracted to youth groups in both rural and urban areas and shall be promoted and supported by active forestry extension activities.

Under the same policy directive, private investors including the landowner groups are also encouraged to actively participate in the reforestation and plantation activities.

Any reforestation programme shall also compliment the Medium Term Development Plan (2011-2015), Papua New Guinea Development Strategic Plan (2010-2030) and the PNG Vision 2050 by establishing forest plantations in addressing 100% down stream processing, climate change issues and poverty in the country.

In order to achieve the above policy and government development plans, provinces are encouraged to plan and establish provincial nurseries and planting programs. Appropriate arrangements will also be made to expand existing plantations and proper management plans develop for these plantations to be commercially viable and self funding.

Currently PNGFA is pursuing a program of improving growing stock in logged over forest under the programme of reforestation naturally. This includes the treatment of logged over forest in major timber project areas to enable timber stand improvement. Land owners will be encouraged to participate and take full ownership in sustainably managing their forest.

Appendix 7a shows the Resource Replacement Programme through reforestation while **Appendix 7b** shows the Reforestation Naturally Programme.

10.3 COMMUNITY FORESTRY

In accordance with the National Forest Policy, PNGFA will develop and maintain community forestry with the view of empowering rural and urban communities in forest related activities such as small-scale sawmilling, eco-tourism, woodlot farming, harvesting and selling of non-timber products to enhance livelihoods and alleviate poverty.

In order to promote landowner participation in the management and utilization of their resources and projects, PNGFA shall:

- Identify opportunities for landowner participation in each project areas.
- Liaise with Provincial Governments (PFMC) and other stakeholders in identifying their provincial needs.
- Identify and conduct appropriate training to enhance capacity building.
- Create a database for all community projects in the country.
- Create a network of extension services with other government agencies, universities/colleges, NGO's and relevant experts to provide guidance and support.

10.4 DOWN STREAM PROCESSING

Domestic processing of forest products is encouraged by the National Government through the PNG Development Strategic Plan (2010-2030) and Medium Term Development Plan (2011-2015) as well as the National Forest Development Guidelines to promote domestic processing with the view of phasing out round log exports over a period of time. The objective of the Government is to increase the level of domestic processing of timber and non timber forest products to about 80% by year 2030.

In order to achieve this PNGFA shall;

- Commercialize existing forest plantations and increase volume for domestic wood processing.
- Conduct feasibility study into the possibility of establishing a Central Wood Processing Facility.
- Review all existing timber permits and project agreements to promote down stream processing of timber.
- Create trade alliances and promote PNG forest resources and wood products both locally and internationally.
- Encourage forest Industries to increase trade volumes of forest produce for domestic and international markets.
- Research, compile and document wood properties of all timber species including lesser known species and upload onto website.
- Promote the marketing and value of lesser known species and non-timber forest products.
- Develop a system to monitor and control the trade of forest products.
- Develop and strengthen the state Marketing Agency (SMA) of PNGFA.

- Enhance standards and quality control for timber treatment, seasoning and grading of sawn timber and other forest produce.
- Regulate all types of domestic processing facilities by setting standards.
- Enhance and strengthen networks with other line agencies.

10.5 CLIMATE CHANGE AND REDD

Forest Accounts for about 17-18% of global carbon dioxide (CO^2) output (UNFCCC). Deforestation has been responsible for almost 90% of the estimated emissions due to land-use change since 1850, with a 20% decrease of the global forest area[3]. It has been estimated also that PNG contributed between 2-7% of the global emissions in 2001 from deforestation and forest degradation[4].

In PNG, the main drivers of CO^2 emissions have been identified to be associated timber harvesting, subsistence agriculture, commercial agriculture, mining, infrastructure development, urban expansion and fire (Shearman, P., et al., 2008).

Various means to mitigate climate change would include afforestation, reduced deforestation and reforestation and others as identified in the Forestry and Climate Change Framework for Action (2009-2015).

PNGFA has established an office of REDD and Climate Change and with collaboration with the FRI (Natural Forest Management Branch) shall oversee and address the issue of climate change. As part of policy initiatives, PNGFA has in place a revised National Forest Development Guide lines (NFDG) and Forestry and Climate Change Framework for Actions 2009-2015 (FCCFA) which will provide a path way to mitigate climate change and green house gasses and adapt to the issues of climate change in the forestry sector in PNG.

In order to address the issue of climate change, PNGFA shall continue to undertake the following:

- Maintain and continue to be active participants in issues relating to climate change, especially Land use Change, Land Use Change and Forestry (LULUCF).
- Maintain and upgrade an information base on issues related to climate change.
- Conduct training and awareness on Climate Change and REDD+ including CDM initiatives to empower relevant stakeholders to manage their environment and forest resources.

[3]Houghton, R.A., 1999: The annual net flux of carbon to the atmosphere from changes in land use 1850-1990. Tellus, **51B**, 298-313.

[4] Shearman,P.,et al.,2008.The State of Forests of Papua New Guinea. Port Moresby: The University of Papua New Guinea.

- Identify areas within the country where possible to initiate REDD+, ARCDM, CDM and watershed management projects.
- Encourage landowners and the timber industries to undertake afforestation and reforestation activities on grassland and degraded sites.
- Continue with research and support other research initiatives into the science of climate change and mitigation.
- Help develop and implement a National Climate Change Policy.
- Help develop and implement a National REDD+ Strategy.
- In collaboration with the Office of Climate Change and Development (OCCD), implement the Climate Compatible Development Plan for PNG.
- Implement the five REDD+ pilot projects as identified by NEC and respective PFMCs to test policy issues.
- Maintain dialogue on the issues of REDD and climate change with all stakeholders at the local, national, regional and the international level.

10.6 RESEARCH

Forestry research is an integral part of the forestry sector. The mandate to control forestry research is set out in Part IV of the National Forest Policy (1990) and the Forestry Act 1991 (as amended) provides the legislative support for implementing this policy.

The PNG Forest Research Institute (FRI) is a directorate of the PNGFA with the broad mission of providing scientific data for the management of PNG's forest resources as well as to provide reputable forest research results and information to PNGFA and other government agencies, private sectors, other research and educational institutions, resource owners and users.

Research is being conducted in collaboration with a number of international agencies such as; JICA, CIFOR, ACIAR and others.

The primary objectives of the institution will be towards;

i. Research into the sustainable management, utilization, conservation and rehabilitation of the nation's natural forest resources.

ii. Screening and domestication of the important indigenous higher value timber species identify suitable species for various specific uses and provide a scientific knowledge base for the establishment and maintenance of plantations for industrial wood as well as conduct research into reforestation and afforestation of the country's large anthropogenic grasslands and degraded lands.

iii. Expand, develop and manage the nationally and internationally important scientific collections as a permanent record of PNG's rich biodiversity and as a resource to enable the key elements of the forest biota to be recognized.

iv. Conduct applied and experimental research into the utilization and marketing of wood and non- wood forest products.

v. Conduct research into climate change, mitigation and adaptation issues.

vi. Conduct research into forest protection (fires, insects and other pathogens).

vii. Collaborate with other PNGFA directorates.

The institute's current structure consists of four (4) research programmes and one (1) support programme aimed at fulfilling the above objectives. The programmes and their specific functions are as follows:

i. **Natural Forest Management.**

To provide the essential growth data needed by forest managers to sustainably manage the nation's natural forest resources, develop and refine silvicutural practices and develop climate change mitigation and adaptation techniques to increase the value and ecosystem services of the forest resources.

ii. **Planted Forest**

To provide a scientific base for the establishment of forest plantations for industrial wood, forest genetic resources, rehabilitation and to improve the germplasm, management and protection of plantations as well as to enhance and maintain insect collections.

iii. **Forest Biology**

Enhance and maintain botanical collections of forest biota and provide mechanisms for identifying plants.

iv. **Forest Products**

To conduct applied and experimental research on wood and non wood properties and their uses as well as provide technical advice and assistance to industries for processing and utilization of their forest products.

v. **Finance and Administration**

To provide support and manage FRI's technical, casuals and professional staff in achieving their goals and objectives and manage the institutes assets, finances and day to day administration.

The research activities are detailed in **Appendix 9** however more infomation on individual programme can be obtained from the Forest Research Institute.

11.0 OTHER FORESTRY RELATED ISSUES

It is acknowledged that there are many other issues that will impact on the forestry sector and within the tenure of this plan or sooner, action will be taken to address issues such as;

11.1 Environment and conservation

The PNGFA is mindful of the need to ensure that any form of timber development activities (felling, road clearance, landing etc) does not endanger the environment. It has in place a Logging Code of Practice (LCOP) and other enabling environmental regulations that are intended to minimize environmental damages.

During the period of this Plan, the PNGFA shall;

- Review the LCOP to ensure compliance and enforce penalties.
- Liaise with relevant Government Agencies, NGO's, Civil Society, Forest Industries and land owners to facilitate environmental concerns in forest development plans.
- Improve the current mapping systems in order to identify and exclude areas exposed to environmental concerns from timber harvesting operations.
- Exclude all proposed conservation areas subject to landowner consent and appropriate legislation dealing with such.

11.2 Non Timber Forest Products

Non Timber Forest Products (NTFP) or sometimes referred to as Minor Forest Products (MFP) is an integral part of any forest ecosystem. Their potential has not been fully utilized over the years as the primary focus has been on commercial forest development.

With the understanding that non timber products can be an important means of providing income for the rural people, PNGFA shall;

- Conduct research and document all known NTFP including carbon.
- Undertake multipurpose NFI and include in the FIMS database.
- Liaise with NGOs and other stakeholders and develop strategies on how best this can be developed to benefit resource owners.
- Explore markets and promote the sales of NTFP.
- Review and develop policies and legislation governing the harvesting and processing of NTFP.
- Conduct awareness programs in rural communities on issues dealing with NTFP and carbon.

11.3 Eco-Forestry

While the primary role of the PNGFA is the sustainable management of the nation's forest resource, it is also appreciative of the other functions of the total ecosystem such as Eco-Forestry. Eco- Forestry (EF) has a number of definitions however can be simplified as "*The sustainable derivation of other benefits from the forest besides traditional timber harvesting*"

EF activities are primarily aimed at generating cash incomes to the local populace through small scale sawmilling activities and harvesting of other forest produce such as medicinal plants, aromatic plants as well as utilizing wildlife like birds and butterflies. Other activities also include preservation of forest ecosystem for eco-tourism and environmental services.

The PNGFA appreciates the need for eco-forestry to be an integral part of forestry development in the country. A unit has been established in recognition of the importance of EF within the organization and shall plan to undertake the following;

- Research all activities that are categorized as EF activities and in consultation with other stakeholders, formulate policy guidelines on how best to utilize and benefit from the different activities associated with EF.
- Formulate strategies and programmes for the enhancement of EF activities.
- Explore markets for EF products and services.
- Collaborate with NGOs, Community Based Organizations (CBO) and other stakeholders and conduct awareness about EF to the landowners.

11.4 Forest Certification

With the global trend on forest certification, PNGFA shall consider developing PNG standards to measure and evaluate the sustainability of our forest. This will in turn enable timber legality, verification, traceability and sustainability to ensure that sustainable forest management practices are being implemented.

Recognizing the fact that forest certification is increasingly becoming a marketing tool and will gain momentum during the tenure of this plan, PNGFA shall;

- Ensure that the officers are well acquainted with the processes of timber certification.
- Develop and promote criteria and indicators as a means of achieving certification in all timber projects.
- Undertake trial projects to test out certification in model forestry projects.
- Engage in international forums and debates regarding certification and timber trade.

12.0 REFERENCES

i. Forest Act 1991 (as amended).
ii. Forest Inventory Mapping Database Reports and Maps.
iii. Forestry Regulations, 1991
iv. Global Forest Resource Assessment, 2005 and 2010
v. Hall,D.O.,J.House, and I.Scrase,2000:An overview of biomass energy
vi. Hammer and J.C Saunders, 1995.Forest Resources and Vegetation Mapping of Papua New Guinea.
vii. Houghton, R.A., 1999: The annual net flux of carbon to the atmosphere from changes in land use 1850-1990.
viii. Internal PNGFA Reports.
ix. National Forest policy, 1991.
x. National Forest development Guidelines, 2009.
xi. Papua New Guinea Forestry Outlook Study, 2009.
xii. Provincial Forest Plans.
xiii. Shearman, P.L., Bryan, J. E., Ash, J., Hunnam, P., Mackay, B. and Lokes, B., 2008. The State of the Forest of Papua New Guinea. Mapping the extent and condition of forest cover and measuring the drivers of forest change in the period 1972-2002. University of Papua New Guinea, 2008.

APPENDIX 1: FOREST RESOURCES OF PNG BY PROVINCES.

Province	Resource as of 1975					Current Resources (2009)		
	Gross Land Area (ha)	Gross Forest Area 1975 (ha)	Adjusted Area (ha)	Gross Volume (m^3)	Converted to Land Use (ha)	Gross Area (ha)	Adjusted Net Area (ha)	Gross Volume (m^3)
Western	9,845,211	6,675,886	5,636,321	173,793,000	1,000,236	5,493,085	4,512,175	143,831,895
Gulf	3,480,065	2,659,466	2,406,954	81,581,500	265,232	2,369,293	2,136,617	73,237,768
Central	2,987,176	2,015,499	1,882,106	87,998,000	290,598	1,665,672	1,570,850	74,654,515
Milne Bay	1,426,403	1,080,043	856,152	29,801,900	170,232	892,895	719,690	25,092,656
Oro	2,277,222	1,726,697	1,571,385	68,734,500	170,450	1,556,247	1,439,135	62,188,786
SHP	2,574,829	2,009,059	1,904,514	92,764,200	133,307	1,875,752	1,817,608	87,875,533
Enga	1,182,384	750,447	723,152	46,084,700	33,856	716,591	696,010	44,458,457
Western Highlands	914,120	509,212	453,805	26,502,500	95,192	414,020	393,403	23,326,118
Jiwaka	-	-	-	-	-	-	-	-
Hela	-	-	-	-	-	-	-	-
Simbu	613,361	393,925	368,183	20,010,400	42,456	351,469	342,253	18,653.153
Eastern Highlands	1,120,510	558,840	530,453	34,868,300	21,679	537,161	512,960	33,687,670
Morobe	3,393,295	2,210,027	2,030,230	109,407,500	246,404	1,947,021	1,809,010	97,202,458
Madang	2,909,527	2,115,612	1,634,487	71,693,900	214,542	1,874,764	1,473,916	65,304,601
East Sepik	381,318	2,103,292	1,886,741	74,839,600	66,098	1,975,616	1,782,228	72,209,530
West Sepik	3,605,390	3,236,371	3,037,119	116,037,800	423,062	2,740,248	2,591,182	99,168,452
Manus	215,029	149,481	105,532	6,250,500	69,395	64,481	47,650	2,985,450
New Ireland	961,034	792,057	728,191	22,466,200	370,588	411,535	381,554	11,638,796
East New Britain	1,534,365	1,312,579	1,223,483	44,465,700	325,221	827,148	779,339	27,887,354
West New Britain	2,045,590	1,814,977	1,696,112	51,848,800	861,501	713,951	684,767	20,206,834
ARB	943,273	708,297	566,349	22,287,600	0	708,297	566,349	22,287,655
Total	**46,410,102**	**32,821,767**	**29,241,269**	**1,181,436,900**	**4,799,849**	**27,136,246**	**24,256,696**	**1,005,897,681**

Source: FIMS Database

Note: Boundaries for Jiwaka and Hela provinces yet to be demarcated in order for FIMS to be updated

Manus	215.029	4.556	0	42.035	90.567	7.009	30.477	1.55	20.400	18.430
N.I.P	961.034	108.005	34.212	155.767	78.599	26.759	91.782	20.20	371.223	74.479
E.N.P	1,534.365	123.095	125.041	210.314	111.601	89.673	247.169	10.15	461.177	156.142
W.N.P	2,045.590	318.212	74.836	221.423	44.097	106.048	103.074	1 9.79	1,004.689	153.418
ARB	943.273	61.111	7.193	0	0	200.565	107.097	55.39	260.849	251.067
Total	**46,410.39**	**4,473.212**	**1,065.470**	**5,813.613**	**3,293.223**	**6,687,810**	**3,959.130**	**1,796,025**	**9,223,602**	**10,098.294**

Source: FIMS Database
Note: No information available for Jiwaka and Hela provinces.

d Hela provinces.

Marshall Lagoon	FMA	01-Apr-2008	10	31-Mar-2018	17,790	14,000	
Varagadi Plantation	FMA	14-Feb-1996	50	13-Feb-2046	6,500	1,280	
Cloudy Bay	FMA	06-Oct-1997	50	05-Oct-2047	148,900	94,013	
					405,432	**341,535**	
Milne Bay							
East Fergurson	TRP	30-Jul-1987	40	29-Jul-2027	31,340	31,254	
East Collingwood	FMA	06-Apr-1998	50	05-Apr-2048	80,700	52,289	
Ulabo Plantation					1,680	1,680	
					113,720	**85,223**	

[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	37,980	37,980	
Aria Vanu Blk 3	LFA		N/A	N/A	39,650	39,650	
Lolo Blk 2 Extn	LFA		N/A	N/A	23,000	23,000	
Andrew Johanna	LFA	26-Feb-1992	N/A	N/A	40,000	40,000	
Rottock Bay	FMA	04-Jan-1996	50	03-Jan-2046	32,739	32,739	Rottock Bay Consolidated
Aria Vanu Blk 2	FMA	06-Apr-1998	50	05-Apr-2048	30,062	30,040	✓
Inland Rauto Miu	FMA	26-Nov-2048	50	25-Nov-2048	101,256	86,126	✓

Manus							
Manus West Coast	TRP	04-Feb-1988	40	03-Feb-2028	32,667	32,667	
West Coast	LFA	05-Dec-1993	N/A	N/A	19,067	19,067	
Jaha	LFA	12-Feb-1993	N/A	N/A	10,111	10,111	
Kali Bay	LFA	14-Mar-1991	N/A	N/A	8,753	8,753	
					70,598	**70,598**	
Jiwaka							New Province
Hela							✓
				TOTAL	**8,303,980**	**6,235,842**	

Source: Forest Resource Acquisition General Information Update, December, 2011.

Note: Total concessions acquired as of November 2011, including those that have expired is 10,972,761 ha.

EHP	512,960	33,687,670	962,505	**385,002**	0	**385.002**	0	0	0	0	0
Simbu	342,253	18,653,153	532,947	**213,179**	0	**213.179**	0	0	0	0	0
WHP	393,403	23,326,118	666,460	**266,584**	0	**266.584**	0	0	0	0	0
Enga	696,010	44,458,457	1,270,242	**508,097**	0	**508.097**	0	0	0	0	0
Jiwaka											
Hela											
Total	**24,256,696**	**1,005,897,681**	**28,739,931**	**10,45,922**	**8,915.257**	**9,876.625**	**7,571 .160**	**7,411.160**	**7,061.160**	**6,746.160**	**6,626.160**

Source: FIMS Database.

Note: No available information for Jiwaka Hela provinces. (New Provinces)

APPENDIX 5b: ANNUAL LEGALLY PERMITTED LOG HARVEST FOR OPERATIONAL PROJECTS

Project Name	Type	TP No	Yr 2013 ('000'm^3)	Yr 2014 ('000'm^3)	Yr 2015 ('000'm^3)	Yr 2016 ('000'm^3)	Yr 2017 ('000'm^3)
Western							
Wawoi Guavi Blk 1,2,3	TRP	TP 01-08	350	350	350	350	350
Makapa	TRP	TP 01-09	170	170	170	170	170
East Awin	FMA	TP 01-01	141	141	141	141	141
Wipim Tapila	FMA		165	165	165	165	165
Semabo	FMA		47	47	47	47	47
			873	**873**	**873**	**873**	**873**
Gulf							
Turama Blk 1	TRP	TP 02-12	216	216	216	216	216
Vailala Blk 1	TRP	TP 02-14	150	150	150	150	150
East Kikori	TRP	TP 02-15	180	180	180	180	180
Turama Extension	FMA	TP 02-12A	340	340	340	340	340
Vailala Blocks 2&3	FMA	TP 02-16	300	300	300	300	300
			1,186	**1,186**	**1,186**	**1,186**	**1,186**
Central							
Lako Imila	LFA	TP 03-01	25	25	25	25	25
Bonua Magarida	TRP	TP 03-28	108	108	108	108	108
Vanapa North	TRP	TP 03-33	100	100	100	100	100
Cloudy Bay	FMA	TP 03-01	61.16	61.16	61.16	61.16	61.16
Marshall Lagoon	FMA	TP 03-36	40	40	40	40	40
			334.16	**334.16**	**334.16**	**334.16**	**334.16**
Milne Bay							
East Fergusion	TRP	TP 04-04	29	29	29	29	29
East Collingwood	FMA		29	29	29	29	29
			58	**58**	**58**	**58**	**58**
Oro							
Yema Gaepa	LFA	TP 05-01	100	100	100	100	100
Ioma Block 4	TRP	TP 05-09	75	75	75	75	75
Ioma Block 5	FMA	TP 05-11	53	53	53	53	53
			228	**228**	**228**	**228**	**228**
Southern Highlands							
East Pangia	FMA	TP 06-01	80	80	80	80	80
			80	**80**	**80**	**80**	**80**
West Sepik							
Bewani	LFA	TP 10-01	80	80	80	80	80
Pegi Pulan	LFA	TP 10-02	80	-	-	-	-
Amanab Blks 1-4	FMA	TP 10-01	100	100	100	100	100
Amanab Blks 5&6	FMA	TP 10-02	108	108	108	108	108
Aitape Lumi Consolidated	FMA	TP 10-03	189	189	189	189	189
			557	**477**	**477**	**477**	**477**
East Sepik							
Hawain	LFA	TP 11-02	150	150	150	150	150
Lower Sepik	LFA	TP 11-01	80	-	-	-	-
April Salume	FMA		167	167	167	167	167
			397	**317**	**317**	**317**	**317**
Madang							
Sogeram	TRP	TP 12-20	150	150	150	150	150
Rai Coast	TRP	TP 12-18	130	130	130	130	130
Middle Ramu Blk 1	FMA	TP-12-01	138	138	138	138	138
Josephstaal	FMA	TP 12-02	132	132	132	132	132
			550	**550**	**550**	**550**	**550**
Morobe							
Umboi	TRP	TP 13-27	60	60	60	60	60

Appendix 5b cont...

Buhem Mongi Busiga	FMA	TP 13-38	61	61	61	61	61
Ongawafa	FMA	TP 13-32	28	28	28	28	28
			149	**149**	**149**	**149**	**149**
West New Britain							
Loko Blk 1	LFA	TP 14-01	80	80	80	80	80
Loko Blk 2	LFA	TP 14-03	80	80	80	80	80
Ania Kapiura Consolidated	TRP	TP 14-52	350	350	-	-	-
Mosa Leim	TRP	TP 14-57	120	120	120	120	-
Kapuluk & Kapuluk Ext	TRP	TP 14-50	160	160	160	-	-
West Arowe	TRP	TP 14-53	155	155	155	-	-
Central Arowe	TRP	TP 14-54	155	155	155	155	155
East Arowe	TRP	TP 14-55	155	155	155	155	155
Atuvo Extension	LFA	TP 14-07	150	150	150	150	150
Anu Alimbit	LFA	TP 14-05	80	80	80	80	80
Andru Johanna	LFA	TP 14-10	100	100	100	100	100
Alimbit Andru	LFA	TP 14-08	80	80	80	80	80
Aria Vanu Blk 3	LFA	TP 14-13	130	130	130	130	130
Aria Vanu Blk 1	LFA	TP 14-06	90	90	90	90	90
Avio Amgen	LFA	TP 14-09	150	150	150	150	150
Passismanua Inland	LFA	TP 14-11	41	41	41	41	41
Passismanua Ext	LFA	TP 14-14	150	150	150	150	150
Rottock Bay Consolidated	FMA	TP 14-02	93	93	93	93	93
Asengseng Consolidated	FMA	TP 14-01	96	96	96	96	96
			2,415	**2,415**	**2,065**	**1,750**	**1,630**
East New Britain							
Open Bay	TRP	TP 15-33	139	139	139	139	139
Inland Pomio	TRP	TP 15-49	93	93	93	93	93
Nutuve	TRP	TP 15-59	120	120	120	120	120
			352	**352**	**352**	**352**	**352**
New Ireland							
Central New Ireland	TRP	TP 16-42	120	120	120	120	120
Danfu	TRP	TP 16-40	60	60	60	60	60
			180	**180**	**180**	**180**	**180**
Manus							
Manus West Coast	TRP	TP 18-08	72	72	72	72	72
Kali Bay	LFA	TP 18-02	60	60	60	60	60
Jaha	LFA	TP 18-01	80	80	80	80	80
			212	**212**	**212**	**212**	**212**
ARB	Nil						
PNG TOTAL			**7,818.160**	**7,652.160**	**7,308.160**	**6993.160**	**6,873.160**

Plan

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Appendix 6 cont...

West Collingwood Bay	182,727	88,912	45,700	150,000	70,000	40,000	40,000	150,000	200,000	**650,000**
	979,542	**503,191**	**258,600**	**300,000**	**203,000**	**112,000**	**112,000**	**450,000**	**600,000**	**1,777,000**
SHP										
Hekiko Upper Kikori	198,711	51,399	26,400	150,000	100,000	50,000	50,000	150,000	200,000	**700,000**
Bosavi	130,809	100,774	51,800	200,000	100,000	50,000	50,000	150,000	200,000	**750,000**
West Giluwe	28,429	12,798	6,500	120,000	20,000	10,000	50,000	150,000	200,000	**550,000**
Nokoli	27,529	1,635	840	110,000	20,000	10,000	10,000	50,000	200,000	**400,000**
Kutubu-Poroma	54,753	11,838	6,000		50,000	25,000	25,000	150,000	200,000	**450,000**
Palopa (Erave)	156,000	156,000	80,200	100,000	100,000	50,000	25,000	150,000	200,000	**625,000**
	596,231	**334,444**	**171,740**	**680,000**	**390,000**	**195,000**	**210,000**	**800,000**	**1,200,000**	**3,475,000**
Enga										
Sau-Taru	165,769	82,856	42,600		25,000	20,000	13,000	150,000	200,000	**488,000**
	165,769	**82,856**	**42,600**	**80,000**	**25,000**	**20,000**	**13,000**	**150,000**	**200,000**	**488,000**
WHP										
Jimi Consolidated	**89,592**	**54,708**	28,100	50,000	25,000	20,000	20,000	100,000	300,000	515,000
	89,592	**54,708**	**28,100**	**50,000**	**25,000**	**20,000**	**20,000**	**100,000**	**300,000**	**515,000**
Simbu										
Karimui Bomai	147,770	59,474	30,500	100,000	50,000	25,000	25,000	150,000	200,000	550,000
	147,000	**59,474**	**30,500**	**100,000**	**50,000**	**25,000**	**25,000**	**150,000**	**200,000**	**550,000**
EHP										
Upper Purari Blks 1&2	80,000	40,000	20,500	100,000	40,000	30,000	50,000	100,000	300,000	620,000
	80,000	**40,000**	**20,500**	**100,000**	**40,000**	**30,000**	**50,000**	**100,000**	**300,000**	**620,000**
Morobe										
Morobe South Coast	169,851	137,871	86,600	50,000	30,000	20,000	20,000	150,000	200,000	**470,000**
Timbe-Kwama	110,948	52,329	32,800	50,000	40,000	20,000	20,000	150,000	200,000	**480,000**
Kwama Masaweng	96,434	46,584	29,200	50,000	40,000	20,000	20,000	150,000	200,000	**480,000**
Mazie Dzia	48,608	38,243	24,000		30,000	20,000	20,000	150,000	200,000	**470,000**
Lake Trist	37,292	32,296	20,300	40,000	30,000	20,000	15,000	150,000	200,000	**455,000**
Umboi Bkl 4	30,000	14,200	8,900	60,000	30,000	20,000	30,000	200,000	200,000	**540,000**
Yagoman	38,191	13,614	8,500	40,000	30,000	20,000	20,000	150,000	200,000	**460,000**
Salamaua	33,318	12,754	8,000	40,000	30,000	20,000	20,000	150,000	200,000	**460,00**
	454,685	**347,891**	**218,300**	**380,000**	**260,000**	**160,000**	**165,000**	**1,250,000**	**1,600,000**	**3,815,000**
Madang										
Ramu Blk 2	98,707	60,002	37,700		40,000	20,000	20,000	150,000	200,000	**430,000**

APPENDIX 7: FCA PROJECTS

PROVINCE/PROJECT	AREA (ha)	FCA No	DATE ISSUED	CURRENT STATUS
Western				
Gre-Dringas Road Project	2,400 (600km)	FCA 01-01	29/11/10	Not operational
Central				
Mekeo Hinterland Integrated Agriculture Project	116,400	FCA 03-01	29/06/09	Not operational. Under dispute
Abeda Integrated Agriculture Project	11,700	FCA 03-02	22/04/10	Active but currently under dispute
Yumi Agro-Forestry Development	115,500	FCA 03-03	03/05/07	Operations suspended due to non compliance.
West Sepik				
Aitape East Integrated Agriculture Project	29,205	FCA 10-01	07/03/08	Operational
Aitape West Integrated Agriculture Project	47,626	FCA 10-02	21/04/08	Operational
Bewani Oil Palm Development Project	139,909	FCA 10-03	26/03/09	Operational
Scotchiao Cocoa Estate Development Project	6,114	FCA 10-04	26/03/09	Operational
East Sepik				
West Turubu Integrated Agriculture Project	121,000	FCA 11-01	26/03/09	Operational
Angoram (Mariemberg) Integrated Ag Project	25,600	FCA 11-02	29/06/09	Operational
East New Britain				
Illi Waswas Roadline	Undefined	FCA 15-01	05/03/07	No progressive report to date.
Illi Waswas Integrated Agriculture Project	38,500	FCA 15-02	05/03/07	Operations suspended.
Illi Standalone Agriculture Project	10,400	FCA 15-03	05/03/07	Expired. New application Lodged.
Inland Lassul Baining Integrated Agriculture proj	30,830	FCA 15-04	15/01/08	Operational
Suikol Makokol Integrated Agricuture	52,000	FCA 15-05	15/12/09	Operational
Mukus Mekoi Integrated Agriculture Proj	68,300	FCA 15-06	07/10/10	Non operational, pending inquiry
Sigta Mukus Integrated Rural Development Project	43,430	FCA 15-07	07/10/07	Operational
New Ireland				
Danfu Integrated Agriculture Project	24,850	FCA 16-01	11/09/09	Operational
TOTAL	**878,764**			

SOURCE: Internal Statistics

APPENDIX 8a: RESOURCE REPLACEMENT PROGRAMME

PLANTATION ESTABLISHMENT					
Province	**Activity**	**Location**	**Species**	**Target (ha/ann)**	**Remarks**
Western			Pometia	250	Medium scale industrial plantation
Gulf			✓	✓	✓
	New plantings	Iokea	Sandalwood	50	Non timber forest plantation
Central	Reforestation	Kuriva	Teak/Mahogany	500	Large scale industrial plantation
	✓	Brown River	Teak	300	Brown River Forest Products
	New plantings	Iva Inika	Sandalwood	50	Non timber forest products
	✓	Kuriva	✓	✓	✓
Milne Bay	Reforestation	Ulabo	Eucalyptus/T.brassii	250	Medium scale industrialplantations
	✓	✓	✓	300	Milne Bay Industries
Southern Highlands	Afforestation	Ialibu	Pinus	100	Small scale industrial plantation
Eastern Highlands	✓	Goroka	Hoop/Klinkii	500	Large scale industrial plantation
	Afforestation	Kainnatu	Pinus	100	Small scale industrial plantation
Western Highlands	✓	Waghi	Eucalyptus	✓	✓
Morobe	✓	Wau/Bulolo	Hoop/Klinkii	250	Medium Scale planatation
	Afforestation	Umi	Hoop/Klinkii	✓	✓
	✓	Leron	Sandalwood	50	
	New plantings		Balsa	40	
Madang	Reforestation	North Coast	Teak/Eucalyptus	250	Medium scale industrial plantation
	✓	Gogol	Acacia/Eucalyptus	1,100	Jant
Madang	New plantings	✓	Balsa	✓	
				40	
East Sepik	Afforestation	Yambi	Teak/Eucalyptus	500	Large scale industrial plantation
	New plantings	Yambi	Eaglewood	50	Non timber forest plantations
	✓	Ambunti	✓	✓	✓
	✓				
West Sepik	Reforestation	Vanimo	Instia/Pometia	500	Large scale industrial plantation
	New plantings	Vanimo	Eaglewood	50	Non Timber Forest plantations
	✓	Bewani	✓	✓	✓
Manus	reforestation	West Coast	Calophyllum	250	Medium scale industrial plantation
New Ireland	✓	Kaut	Calophyllum	100	Small scale industrial plantation
East New Britain	✓	Movelo	Teak/Eucalyptus	500	Large scale industrial plantation
	Reforestation	Kerevat	Teak	100	Small scale industrial plantation
	New plantings	Pomio	Balsa	✓	
	✓	Kerevat	✓	40	
	✓			40	
West New Britain	Reforestaion	Kandrian	Homalium	250	Medium scale industrial platation
	✓	Buluma	Eucalyptus//Teak	750	Stettin Bay Lumber Co
Bougainville	✓	✓	✓		
	✓		✓	✓	
	✓		✓	✓	
	✓		✓	✓	
Province	**Activity (Nursery)**	**Location**	**Species**	**Annual Target**	**Remarks**
Western		Malam	Sandalwood	40,000	Non-Timber
Gulf		Iokea	"	✓	✓
Central		Kuriva	Teak	320,000	
		✓	Mahagony	80,000	
		✓	Sandalwood	40,000	Non-Timber
		Hohola	Exotic/Mix spp	10,000	Extension
Milne Bay		Ulabo	Eucalyptus	16,000	
		✓	T.brassii	40,000	
		✓	Dracontomelon	✓	
		Kiriwina	Mix indigenous spp	10,000	Extension
Oro		Popondetta	Local spp	✓	✓
Southern Highlands		Ialibu	Pinus	✓	
			Nothafagus	40,000	

Appendix 8a cont...

		Mendi	Casuarina	5,000	Extension
		✓	Eucalyptus	✓	
Enga		Laiagam	Casuarina	20,000	
Western Highlands		Waghi	Eucalyptus	40,000	
		✓	Hoop/Klinkii	✓	
		Kagamuga	Casuarina	50,000	
Chimbu		Kundiawa	Mix species	30,000	
Eastern Highlands		Lapegu	Pinus spp	90,000	
		✓	Hoop/Klinkii	80,000	
		Fayantina	Pinus spp	90,000	
		Kainantu	Pinus spp	50,000	
		✓	Hoop	40,000	
Morobe		Umi (Lae)	E.pelita	120,000	
		✓	Pinus	18,000	
		Wau/Bulolo	Hoop	80,000	
		✓	Klinkii	✓	
		✓	Agathis	40,000	
		✓	Pinus spp	✓	
		Oomsis	Mix species	10,000	
		Leron	Sandalwoo/Pinus	40,000	Non-Timber
Madang		North Coast	Eucalyptus	120,000	
		✓	Teak	✓	
		Baku (Gogol)	Acacia mangium	10,000	Extension
East Sepik		Yambi	Eucalyptus	80,000	
		✓	Teak	160,000	
		✓	Klinkii	80,000	/
		✓	T.brassii	✓	
		✓	Eaglewood	✓	
		Ambunti	"	40,000	
		Wewak	Balsa	10,000	
West Sepik		Bewani	Instia	240,000	
		✓	Pometia	80,000	
		✓	Hopea	✓	
		✓	Eaglewood	✓	
Manus		West Coast	Calophyllum	40,000	
		✓	Dillenia	✓	
New Ireland		Kaut	Calophyllum	✓	
		✓	Eucalptus	✓	
East New Britain			Teak	240,000	
			Pometia	80,000	
			Calophyllum	✓	
			Balsa	50,000	
West New Britain		Kandrian	Homalium	120,000	
		✓	Calophyllum	✓	
Bougainville			Balsa & mix species		Extension
			Albizzia		✓
					✓

SOURCE: Provincial Forest plans

	2017
	(K)
	45,000
	45,000
	45,000
	45,000
	45,000
	45,000
	45,000
	45,000
	20,000
	45,000
	45,000
	470,000

APPENDIX 8b: NATURAL FOREST MANAGEMENT

YEAR AND BUDGET ESTIMATES (Kina)			
Province	Project name	Area (ha)	2017
			(K)
Central	Cloudy Bay	2,500	45,000
West Sepik	Vanimo Blocks	2,500	45,000
	Amanab 1-4	2,500	45,000
	Amanab 5-6	2,500	45,000
	Altape Lumi	2,500	45,000
Madang	Ramu Block 1	2,500	45,000
	Rai Coast	2,500	45,000
	Sogeram	2,500	45,000
Manus	West Coast	1,000	20,000
W.N. Britain	Asengseng	2,500	45,000
	Rotfock Bay	2,500	45,000
	TOTAL	27,000	470,000

APPENDIX 9: RESEARCH PROGRAM

PROGRAM NAME/OBJECTIVES	ACTIVITY DESCRIPTION	EXPECTED TARGET		ESTIMATED COST(K)	
1.0 Forest Biology		2012	2015	2012	2015
Plant species for future benefits	Compile species checklist for Permanent Forest Estates	Field work to targeted projects completed	Technical report for each site available for review	50,000	50,000
	Develop key to timber trees of Morobe	Field work completed for targeted areas	50% of timber tree species covered	40,000	40,000
	Taxonomic revision of Syzgium, Diospyros, Dillenia, Polyosma	Botanical literature completed	Draft manual completed and available for review	10,000	10,000
	Determine the ecology of the natural low diversity forests for the management of planted forest	Field work completed for two sites	Report drafted and submitted	30,000	30,000
	Develop educational awareness on biodiversity and its conservation	Conduct awareness sessions to the communities & stakeholders	Awareness program in place	25,000	25,000
Ex-Situm Conservation	Maintenance of botanical infrastructures	Fencing of botanical gardens, maintenance of bridges, & sealing of internal roads.	Perimeter fence in place, bridges maintained & upgraded and improve image of the botanical gardens	40,000	40,000
	Orchid nursery redevelopment	Maintenance of shed houses, germination sheds, beds, water system & fence	Improve facilities in the nursery	20,000	20,000
	Plant collection propagation and development	Increase stockings of horticulture plants in the gardens	Increase sales of plants to the public	15,000	15,000
	Maintenance of capital and operational assets	Maintenance of capital assets	Equipment in operating conditions and materials in stock.	20,000	25,000
Maintain biological collections	Database herbarium and botanical gardens plant collection	New records entered into the database	Plant database available online	5,000	5,000
	Curate and manage herbarium collections	Curate incoming collections and update revise groups	Well maintained collected & updated records & information	10,000	10,000
	Educational display of horticultural, rare, endemic & threaten species	Documentation of horticultural, rare, endemic, threaten and invasive species.	Increase public awareness on flora.	20,000	20,000
	Orchid research and conservation	Documentation of new orchid records and species	Better understanding of orchids stocks in the botanical gardens	30,000	30,000
Promote in-situ Biodiversity conservation	Establishment of biodiversity plots in natural forest	2 PSP plots established in natural forest	Plots established and data analyzed	20,000	20,000
	Forest inventory of threaten forest habitats	Preliminary survey and assessment of threaten habitats	Updated list of threaten forest habits in PNG	25,000	25,000
	Species recovery and rehabilitation of degraded forest habitats	Determine and investigate threaten species	Strategies for recovery and rehabilitation prepared.	15,000	15,000

			included for supporting climate change.	10,000	20,000
	Undertake ecological studies on productivity and suitability of rehabilitation sites	Conduct soils and vegetation studies	Gathered information publish	6,000	6,000
Improve Silviculture techniques	Development of Silviculture techniques for stand improvement	Secure demonstration sites in Vanimo	Establishment of 50 hectares and report baseline assessment and treatments	100,000	80,000
	Tree planting in Silvicultural demonstration trials in Madang and Vanimo	Gap planting with local high value commercial species	Improve stocking of preferred local species	100,000	25,000
	Production of Silvicultural prescriptions manuals	Undertake review and conduct seminars	Update manuals with new data from trial plots.	50,000	50,000
Studies on logging impacts, nutrient budgets and forest watershed and soils of PNG	Monitoring of forest changes through assessment in the ecological plots	Annual plot measurement, data entry and analysis.	Publish results of 15 years of monitoring of forest changes in Finchafen	20,000	26,00
	Undertake soil surveys in PSPs, plantation, research trials, and REDD pilot sites	Completed chemical analysis tests	Drafting of nutrient status reports.	100,000	300,000
	Forest biomass sampling in collaboration with ACP EU project	Destructive sampling of forests	Develop biomass algometric equations for 20 species	200,000	300,000
	Assessing logging impact on soil and water quality	Undertake field studies	Draft evaluation report for logging code of practice	50,000	4,000
	Undertake nutrient analysis from soil, water and plants.	Complete tests of all prepared samples	Compile report on Chemical analysis	10,000	10,000
	Undertake forest health assessment in plots from plantations, logged and unlogged forest	Compile forest health and Performance report	Document results of evaluations	5,000	5,000
	Survey of forest for consideration of water	Establish water quality and quality	Establishment report compile		

Appendix 9 cont...

	protection	measurement trials		40,000	40,000
	Evaluate land suitability classification for uses	Complete reports from soil and vegetation relationship analyses	Secure research sites, mapping of plots and publish papers.	5,000	5,000
Improve logging and harvesting techniques (RIL)	Test forest growth simulation model, PINFORM in impact areas	Application in REDD pilot projects	Propose changes to harvesting regimes and increases usage of the model	20,000	26,000
	Development of forest growth model	Data validation and modeling	Present model	10,000	20,000
	Assessment of reduced impact logging	Monitor harvesting in three sites	Compile report	20,000	30,000
Provide data on forest biomass and soils for carbon accounting for REDD and trade options (Climate Change)	Conduct inventory on forest carbon in collaboration with the Mapping and Inventory branch	Complete survey of five REDD pilot projects and three impact areas	Compile data on carbon stock for the REDD project and impact areas and information stored in the database	200,000	100,000
	Conduct awareness on climate change development of REDD projects	Secure sites for conservation and potential REDD+	Mitigation actions supporting low carbon economy	100,000	300,000
	Provide information to Climate Change Office	Complete GHG annual report and submit	Publish papers on GHG emission from forestry sector	100,000	100,000
Assessment of forest vulnerability and adaptation for climate change impacts	Conduct vulnerability and adaptation survey on forest sites threaten by climate change	Complete survey of four identified threaten sites	Compile data and provide report to decision & policy makers	80,000	60,000
	Establishment of permanent reference points for future monitoring	Establish reference [points for monitoring changes	Provide report on the impacts of climate change on the forest and peoples livelihood	100,000	300,000
	Provide information to climate change office	Sector's V&A report submitted	Papers published and information made available	50,000	30,000
	Mapping of forest areas under threat of climate change	Integrate information into PNGFA's mapping system	Documents are publish and maps are made available	50,000	40,000
	Documentation of resilience on threaten forest sites to affected by climate change	Documentation of affected areas	Reports publish on risks and disaster areas	40,000	40,000
Provide data and information on sector's GHG emissions	Mapping and computation of carbon stock changes from forestry and land use changes	Reports completed and documented	GHG data compile and reports made available on carbon pools of PNG	80,000	60,000
	CDM carbon stock assessments	Establish field trials	Compile establishment report and monitor enhance carbon stock	50,000	30,000
	Bio-energy fuel wood survey	Survey methods accepted and data collected	Report on carbon values of PNG species publish and make available	50,000	40,000
Development of community REDD and CDM (Environmental Services)	Computation of carbon stock changes from reforestation and Agro-forestry	Reports completed and documented	Data on PNG's carbon stock changes compile and reports made available	80,000	60,000
3.0 Planted Forest					
Identify potential	Conduct research to determine the possibility of	Develop project proposals for Eaglewood	Develop techniques for inoculation		

	Develop Balsa SSO at Kerevat	Collect seeds and stored, ready for distribution	Develop techniques to improve production of quality seeds and identify sites for new seed orchards	8,000	32,000
	Identify and protect Teak seed trees and progeny trial in Kerevat and Kuriva plantation	Collect seeds and stored, ready for distribution.	Develop techniques to improve production of quality seeds and identify sites for new seed orchards	8,000	32,000
Acquire land for the establishment of new Seed Orchards	Formalize 200 ha land at Open Bay for establishment of seed production for Lowland plantation species	Dialogue with Forest Development Directorate (FDD) and Open Bay Project staff.	Seedling and clonal of Teak,Balsa, Kamarere, E.pellita &T.brassii with the target of increasing to 20 species in future	12,000	48,000
	Formalize land at Bulolo for the establishment of seed production for Pinus spp, Hoop and Klinkii	Dialogue with Bulolo Plantation Project and FDD	Establishment of SSO with the target of increasing to 10 higher altitude plantation species in future.	10,000	40,000
	Formulate agreement with landowners of Bumsi.Mereand Markham bridge for use of land	Dialogue with respective landowners	Formal agreement of 5 ha with intention of increasing to 20 ha in		

	Develop procedures for supply of quality seeds to customers	Create listings of all customers from both within and abroad	Develop standard procedures, pricing and registry of all customers	10,000	10,000
	Develop appropriate techniques for seed germination for each species.	Test for optical germination, viability & moisture content	Develop standard techniques for major plantation species and publish information	5,000	10,000
	Develop mechanisms for collecting large quantity of seeds from within and abroard	Purchase Acacia spp & Sandal wood seeds from Western Province	Collect 500 kg and produce a collection report	250,000	300,000
Plantation establishment (Matching species to site)	Develop pilot plantations of potential Lowland species of native species	Select sites in Vanimo & establish 10 ha of Kwila & 10 ha of Taun in Open Bay	Establish 10 ha of E.pellita at Open Bay, 10 ha of Mersawa at Lae with the intention of introducing 10 potential species	20,000	100,000
	Develop pilot plantations of indigenous species of the Highlands	Select sites and negotiate with landowners	Establishment of 10 ha trial plantation	15,000	50,000
	Develop nursery practices for indigenous species	Identify species and test Silvicultural techniques	Develop techniques for 100 species of commercial interest and publish report	5,000	15,000
	Develop appropriate silvicultural techniques for potential native species of PNG	Produce Silvicultural manual for Taun, Kwila, E.pellita &Calophyllum	Develop manuals for Malas, Wau Beech, Basswood & in future for 50 more species.	40,000	80,000
Improve management of existing plantation	Develop management plan for plantations in PNG	Draft plans for Ulabo and Fayantina projects	Review of Bulolo,Ulabo & Fayantina and FDD to produce plans for new projects	15,000	30,000
	Develop techniques to control termite in Bulolo Plantations	Assess progress of termite control trials	Control measures identified and results publish	10,000	30,000
	Research into improving the current silvicutural practices.	Identify problems in current practices.	Establish trials to address problem and produce reports.	10,000	10,000
	Development of Growth and Yield and volume tables for current plantation species	Collect data for Teak (Madang) and develop Teak volume table	Complete volume table tables for other plantation species	10,000	30,000
	Develop Silvicultural manual for major plantation species in PNG	Collect require data for development of Silvicultural manual for E.degulpta and Teak	Collect data and revise Silvicultural techniques for Hoop, Klinkii, P. carribaea and other plantation		

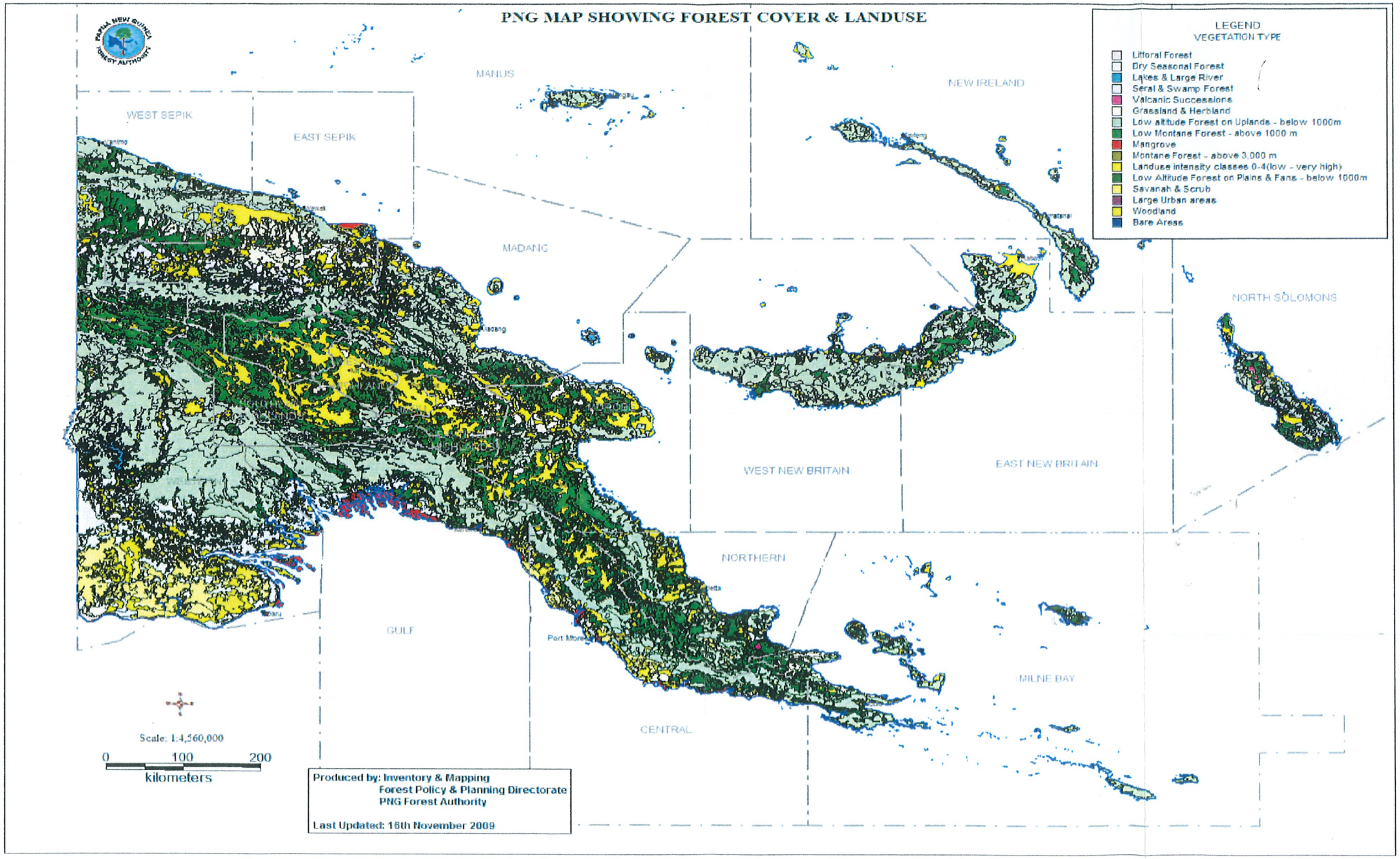
Appendix 9 cont...

			Species in PNG	10,00	30,000
	Develop appropriate techniques for growing hardwood species in the grassland areas of PNG	Assess growth of hardwoods under different shade or nurse crops in Umi and Leron	Identify technique for growing hardwoods in grassland for future large scale afforestataion programmes	5,000	20,000
Forest protection (Pest, Disease and Fire)	Develop appropriate techniques to assess and control termite attacks in Bulolo, Umi,and Leron projects	Establish plots, and apply control measures	Identify control measures and include in Umi project management plan	6,000	18,000
	Develop appropriate control techniques to assess and control disease attack on E.deglupta in PNG	Establish plots in Ulabo	Identify diseases and control measures and include in management plan	10,000	30,000
	Develop appropriate techniques to assess and control termite attack on tree species in the Botanical Gardens	Survey and identify termite location points	Identify control measures and include in the Management plan	10,000	30,000
	Develop appropriate techniques for identification and collection of insects for storage in the insect lab at FRI	Collect, label all insects and enter details into insect data base.	Produce report.	5,000	15,000
	Develop appropriate insect database at FRI	List all collected insect species	Update database and produce report	1,000	5,000
	Review and assess forest fires in plantations and natural forests	Review past work on forest fires in PNG	Develop fire control measures for plantation and natural forest in PNG	15,000	45,000
Plantation for fuel wood and carbon trade options	Identify species for fuel wood and charcoal productions in PNG	Harvest tress from current fuel wood trials in Central and Western Highlands	Identify species, develop techniques and demonstrate carbon productions to local communities.	10,000	30,000
4.0 Forest Products					
Wood properties and identification	Investigate timber drying properties, and determine schedules for plantation and other species	Setup of lab and equipment , collections of samples, testing in kiln dryer	Determine drying schedules and report findings to industries	13,000	8,000
	Determine physical and mechanical properties of wood (Wood properties)	Sample preparation and testing of species in the Lab.	Number of properties determine for a particular species.	35,000	30,000
	Wood identification and study of anatomical properties	Samples identified, special features observed and entered into database	Database updated and uploaded onto FRI website.	25,000	20,000
Preservation	Undertake decay and termite studies(Durability studies)	Sample preparation and conduct studies	Number of plots establish under different environment for a given species	20,000	20,000
Promotion of timber marketing information	Disseminate information on wood drying and properties to Industries as well as to NFS	Data compiled and promoted by the marketing branch (MkIS)	Information is available in MkIS .	5,000	3,000 5,000
Assessment and	Create database of existing wood processing	Survey of static mills in the country and	Database system established and		

Appendix 9 cont...

Improvement of sawmills (Wood Processing)	standards	Creation of a database system	reports produce for policy change in regard to down stream processing	10,000	5,000
Product Developments (Standards)	Review and promote wood processing standards	Collect information, disseminate and form working group.	Working group established and relevant institutions consulted.	15,000	15,000
Investigate solid wood manufacture (Product Development)	Contact and seek collaborating partners	Interest shown by stakeholders and collaborative work undertaken.		5,000	5,000
			TOTAL	**8,327,000**	**5,187,000**

FOREST COVER AND LANDUSE MAP



TIMBER CONCESSION MAP OF PNG

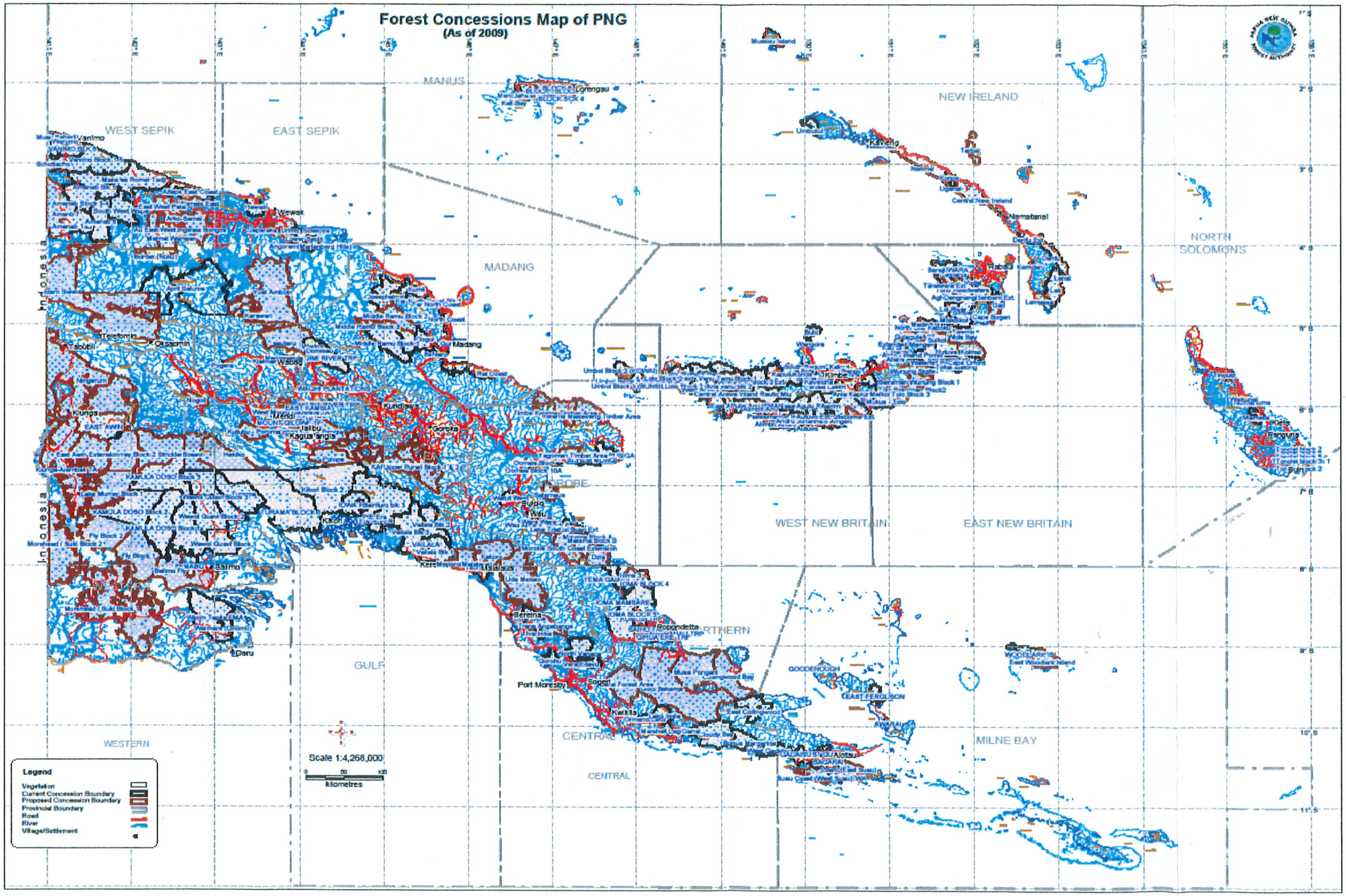


TABLE 3: CONCESSION TABLE

PLAN ID	CEN PROV	EXP_CUR	CONSTYPE	STATUS	611	Collingwood Bay			Proposed	
313	Ome Ome	Expired	TRP	Concession	612	Musa Pongani			Proposed	
315	Marshall Lagon	Current	FMA	Concession	614	Ioma 3			Proposed	
314	Varagadi	Current	FMA	Concession	613	Goro Itakama			Proposed	
316	Cloudy Bay	Current	FMA	Concession	**PLAN ID**	**GUL PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	
305	North Vanapa	Current	TRP	Concession	210	Vailala Block 2	Current	FMA	Concession	
312	Edevu	Expired	TRP	Concession	208	East Kikori	Current	TRP	Concession	
304	Ganai	Expired	TRP	Concession	209	Tauri Meporo	Current	TRP	Concession	
303	Iva Inika	Current	TRP	Concession	206	Turama	Current	TRP	Concession	
306	Bonua Margarida	Current	TRP	Concession	211	Turama Block 1	Current	FMA	Concession	
309	West Gadaisu	Current	TRP	Concession	217	Vailala Block 3	Current	FMA	Concession	
307	Gorohu	Current	TRP	Concession	207	Vailala	Current	TRP	Concession	
308	Ormand Lako	Current	TRP	Concession	203	Eia River	Expired	TRP	Concession	
311	Lako Imila	Current	LFA	Concession	204	Pie River	Expired	TRP	Concession	
323	Uda Mekeo		PFD	Proposed	205	Iowa River	Expired	TRP	Concession	
321	Trans Angabanga		PFD	Proposed	124	Kakoro			Proposed	
325	Barai Forest Area			Proposed	215	Meporo (Malalaua)			Proposed	
PLAN ID	**MIL PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	212	Kikori Block 2	Current	FMA	Concession	
507	West Gurney	Expired	TRP	Concession	218	Vailala Block 1	Current	TRP	Concession	
504	Gadaisu	Current	TRP	Concession	213	Baimuru Block 3	Current	FMA	Concession	
503	Sagarai	Expired	TRP	Concession	**PLAN ID**	**NIR PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	
506	Gara Modewa	Expired	TRP	Concession	1706	Mussau Island	Expired	TRP	Concession	
508	Woodlark Is	Expired	TRP	Concession	1715	Lenai	Expired	TRP	Concession	
502	Awarai	Expired	TRP	Concession	1713	Danfu	Current	TRP	Concession	
505	East Fergusion	Current	TRP	Concession	1714	Danfu Ext	Expired	TRP	Concession	
501	GoodEnough	Expired	TRP	Concession	1719	Uganai	Expired	TRP	Concession	
511	East Collingwood	Current	FMA	Concession	1705	Nakmai	Current	LFA	Concession	
516	West Suau		PFD	Proposed	1720	West Kaut	Expired	TRP	Concession	
513	Loani (East Suau)		PFD	Proposed	1718	Umbukul	Current	TRP	Concession	
512	East Woodlark Is		PFD	Proposed	1701	Kaut	Expired	TRP	Concession	
PLAN ID	**ORO PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	1712	Central New Ireland	Current	TRP	Concession	
609	Yema Gaiapa	Current	LFA	Concession	1716	Lak	Expired	TRP	Concession	
608	Embie Hanau	Expired	TRP	Concession	1708	Konos	Expired	TRP	Concession	
607	Girua Ere	Expired	TRP	Concession	1723	Lamassa		PFD	Proposed	
603	Saiho	Current	TRP	Concession	1702	Kamdaru	Expired	TRP	Concession	
604	Ioma Block 4	Current	TRP	Concession	1724	Tabar	Current	LFA	Concession	
601	Ioma Mambare	Current	TRP	Concession	1934	Bango		TRP	Concession	
602	Kumusi		TRP	Concession	1931	Alolo		TRP	Concession	
610	Ioma Block 5	Current	FMA	Concession	1932	Kau'uli	Expired	TRP	Concession	

PLAN ID	WNB PROV	EXP_CUR	CONSTYPE	STATUS	1909	Ibana	Expired	TRP	Concession
1912	Mosa Leim	Expired	TRP	Concession	1910	North Ulamona	Expired	TRP	Concession
1952	Pasismanua Inland	Current	LFA	Concession	1925	Madedua	Expired	TRP	Concession
1953	Passismanua Ext	Current	LFA	Concession	1924	Ibana East	Expired	TRP	Concession
1923	Ulamona Ext	Expired	TRP	Concession	1927	Gulagula	Expired	TRP	Concession
1930	Hargy	Expired	TRP	Concession	1926	Extended Mengen	Expired	TRP	Concession
1933	Sisimi	Expired	TRP	Concession	1928	Aliiai	Expired	TRP	Concession
1904	Uletevasa	Expired	TRP	Concession	1929	Ania Kapiura	Expired	LFA	Concession
1902	Witori	Expired	TRP	Concession	1954	Andru Johanna	Current	LFA	Concession
1901	Mami	Expired	TRP	Concession	1949	Autuvo	Current	LFA	Concession
1905	Malalimi	Expired	TRP	Concession	1958	Asengseng	Current	FMA	Concession
1920	South Bagum	Expired	TRP	Concession	1960	Vanu Tamu	Current	FMA	Concession
1962	Mosa	Expired	TRP	Concession	1964	Agulu Reserve	Current	FMA	Concession
1918	Kisang	Expired	TRP	Concession	1963	Asirim	Current	FMA	Concession
1915	Malum	Expired	TRP	Concession	1961	Aria Vanu Blk 2	Current	FMA	Concession
1916	Bunga	Expired	TRP	Concession	1935	Kapuluk	Current	TRP	Concession
1914	Toa	Expired	TRP	Concession	1999	Inland Rauto Miu	Current	FMA	
1913	Keveloha	Expired	TRP	Concession	**PLAN ID**	**MAD PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
1944	Lolo Blk 1	Current	LFA	Concession	1309	Sogeram	Current	TRP	Concession
1957	Rottock Bay	Current	FMA	Concession	1312	Josephstaal	Current	FMA	Concession
1922	Bulu	Expired	TRP	Concession	1301	Gogol	Expired	TRP	Concession
1908	Wangore	Expired	TRP	Concession	1302	Gum	Expired	TRP	Concession
1955	Lolo Blk 2 Ext	Current	LFA	Concession	1308	Barum	Expired	TRP	Concession
1943	Lolo Blk 2	Current	LFA	Concession	1307	Kumil	Expired	TRP	Concession
1945	West Arawe	Current	TRP	Concession	1310	Far North Coast Blk 3	Expired	TRP	Concession
1947	Central Arawe	Current	TRP	Concession	1305	Far North Coast	Expired	TRP	Concession
1948	East Arawe	Current	TRP	Concession	1304	North Coast	Expired	TRP	Concession
1939	Pulie Anu	Current	LFA	Concession	1303	Naru	Expired	TRP	Concession
1941	Alimbit Anu	Current	LFA	Concession	1311	Rai Coast	Current	TRP	
1946	Alimbit Andru	Current	LFA	Concession	1315	Middle Ramu Blk 3		PFD	Proposed
1951	Awio Amgen	Current	LFA	Concession	1314	Middle Ramu Blk 2		PFD	Proposed
1921	Fulleborne Ext	Expired	TRP	Concession	1313	Middle Ramu Blk 1	Current	FMA	Concession
1911	Kapuluk Ext	Expired	TRP	Concession	**PLAN ID**	**WHY PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
1956	Aria Vanu Blk 3	Current	LFA	Concession	901	Kum NO 2	Expired	TRP	Concession
1942	Aria Vanu Blk 1	Current	LFA	Concession	902	Kuna River	Expired	TRP	Concession
1919	Bola	Expired	TRP	Concession	916	Waghi Plantations	Expired	PFD	Proposed
1917	Honde-Laulimi	Expired	TRP	Concession	915	East Kambia	Expired		Proposed
1906	Galae	Expired	TRP	Concession	904	Jimi River	Expired	TRP	Concession
1903	Kako	Expired	TRP	Concession	**PLAN ID**	**EHY PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
1934	Bango	Expired	TRP	Concession	1101	Upper Purari Blk 1&2			Proposed

PLAN ID	MOR PROV	EXP_CUR	CONSTYPE	STATUS	1508	Amanab Block 3-4	Current	FMA	Concession
1213	Watut West	Expired	TRP	Concession	1509	Aitape East Coast	Current	FMA	Concession
1204	Wau Area 4 Blk 4	Current	TRP	Concession	1507	South West Wapei	Current	FMA	Concession
1211	Oomsis Blk 10 A	Expired	TRP	Concession	1511	Au East Coast			Proposed
1201	Oomsis Blk 2	Expired	TRP	Concession	1512	East west Wapei			Proposed
1203	Were Were	Current	TRP	Concession	1517	Schotiacho			Proposed
1208	Kui Buso	Expired	TRP	Concession	1510	Arko-Samei			Proposed
1212	Kui Buso Ext	Expired	TRP	Concession	1514	Maimai Wanwan			Proposed
1210	Maiama Blk A	Expired	TRP	Concession	1513	Idam Siawi			Proposed
1202	Maiama Blk B	Expired	TRP	Concession	1524	Amanab 1&2	Current	FMA	Concession
1209	Morobe Coast	Expired	TRP	Concession	1525	Amanab 7			Proposed
1207	Umboi Blk 3(Kaimanga	Current	TRP	Concession	1522	Amanab 6	Current	FMA	Concession
1206	Umboi Blk 2 (Kowai)	Current	TRP	Concession	1521	Amnab Blk 5	Current	FMA	Concession
1205	Umboi Blk 1 (Bunsil)	Current	FMA	Concession	1516	Wes Romei Tadji	Current	FMA	Concession
1214	Mongi Busiga	Current	FMA	Concession	1526	Make			Proposed
1215	Buhem Mongi	Current	FMA	Concession	**PLAN ID**	**SHP PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
1217	Dzia		PFD	Proposed	707	Rongo Mondane	Current	TRP	Concession
1218	Lake Trist		PFD	Proposed	708	Koia Wekomini	Expired	TRP	Concession
1219	Salamaua		PFD	Proposed	702	Mount Giluwe 1B	Expired	TRP	Concession
1231	Timbe Kwama TA			Proposed	704	Mount Giluwe 1C	Current	TRP	Concession
1232	Kwama Masaweng TA			Proposed	705	Mount Giluwe 1D	Expired	TRP	Concession
1228	Yagoman Timber Area			Proposed	706	Mount Giliwe 1E	Expired	TRP	Concession
1230	Morobe South Coast Ext			Proposed	715	Mount Giluwe 1F	Expired	TRP	Concession
PLAN ID	**ESK PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	703	Kalgabunbu	Expired	TRP	Concession
1403	Hawain	Current	LFA	Concession	701	Mount Giluwe 1A	Expired	TRP	Concession
1404	Lower Sepik	Current	LFA	Concession	714	West Mount Giluwe			Proposed
1405	April Salume	Current	FMA	Concession	713	Nogoli			Proposed
1401	Angoram (Mariemberg)	Expired	Expire	Concession	712	Kutubu Poroma			Proposed
1410	Yuat Amboin		TRP	Concession	710	East Pangia			Proposed
1407	Japaraka-Urimo-Sosoya			Proposed	711	Hekiko			Proposed
1406	Nungwaia Bongos		PFD	Proposed	709	Bosavi			Proposed
1413	Border (Nuku)		PFD	Proposed	715	Polopa			Proposed
1414	Karawari		PFD	Proposed	**PLAN ID**	**SIM PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
1415	Wamsis East		PFD	Proposed	1004	Karamui Bomai			Proposed
PLAN ID	**WSK PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	**PLAN ID**	**ENG PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
1505	Bewani	Current	LFA	Concession	803	Waripa	Expired	TRP	Proposed
1506	Pegi Pulan	Current	LFA	Concession	802	Domisau	Expired	TRP	Proposed
1501	Musu Saberjin	Expired	TRP	Concession	899	Sautaru			Proposed
1502	Waromo	Expired	TRP	Concession	805	Pogera	Expired	TRP	Proposed
1504	Vanimo Block 1-5	Expired	TRP	Concession	801	Kiaimanda	Expired	TRP	Proposed

1503	Vanimo Block 6	Expired	TRP	Concession					
PLAN ID	**NSL PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	**PLAN ID**	**MAN PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
2003	Tonolei Block 1	Expired	TRP	Proposed	1602	Manus West Coast	Current	TRP	Concession
2008	Tonolei Block 6	Expired	TRP	Proposed	1603	Jaha	Current	LFA	Concession
2007	Tonolei Block 5	Expired	TRP	Proposed	1604	Kali Bay	Current	LFA	Concession
2006	Tonolei Block 4	Expired	TRP	Proposed	1609	Block 7			Proposed
2005	Tonolei Block 3	Expired	TRP	Proposed	1608	Block 6			Proposed
2004	Tonolei Block 2	Expired	TRP	Proposed	1607	Block 5			Proposed
PLAN ID	**ENB_PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**	1606	Block 4			Proposed
1819	Senbem Ext	Expired	TRP	Concession	1605	Block 3			Proposed
1812	Ramandu	Expired	TRP	Concession	**PLAN ID**	**WES_PROV**	**EXP_CUR**	**CONSTYPE**	**STATUS**
1822	Bergberg Ext	Expired	TRP	Concession	105	Makapa	Current	TRP	Concession
1817	Nutuve	Expired	TRP	Concession	101	Waimare (Oriomo)	Expired	TRP	Concession
1814	Gar	Expired	TRP	Concession	104	Wawoi Guavi Blk 3	Current	TRP	Concession
1820	Waterfall Bay Ext	Expired	TRP	Concession	102	Wawoi Guavi Blk 1	Current	TRP	Concession
1811	Gaulim Ext	Expired	TRP	Concession	103	Wawoi Guavi Blk 2	Current	TRP	Concession
1815	Inland Pomio	Expired	TRP	Concession	106	Semabo	Current	FMA	Concession
1813	Tokai Matong	Expired	TRP	Concession	108	Wipim Tapila	Current	FMA	Concession
1821	Cape Orford	Expired	TRP	Concession	107	East Awin	Current	FMA	Concession
1804	Tamoip	Expired	TRP	Concession	113	Kamula Doso Blk 2	Current	FMA	Concession
1803	Simbali	Expired	TRP	Concession	112	Kamula Doso Blk 1	Current	FMA	Concession
1806	Aghathat	Expired	TRP	Concession	114	Kamula Doso Blk 3	Current	FMA	Concession
1816	Aghathat Ext	Expired	TRP	Concession	116	Nomad Strickland		PFD	Proposed
1807	Dengnangi	Expired	TRP	Concession	111	Fly Block 2		PFD	Proposed
1801	Kaboku	Expired	TRP	Concession	115	Lake Murray Block 1		PFD	Proposed
1805	Loi River	Expired	TRP	Concession	117	Lake Murray Block 2		PFD	Proposed
1802	Molkolkol	Expired	TRP	Concession	110	Fly Block 1		PFD	Proposed
1809	Vudal	Expired	TRP	Concession	109	Balimo Fly		PFD	Proposed
1823	Mukus-Tolo	Current	FMA	Concession	120	Morehead/Suki Blk 1		PFD	Proposed
1818	Taraiwara	Expired	TRP	Concession	121	Morehead/Suki Blk 2		PFD	Proposed
1835	Wunung Sigite			Proposed	119	Ningerum		PFD	Proposed
1828	Bairaman Wunung Blk1			Proposed	122	East Awin Extension			Proposed
1829	Tolo Bairaman Blk 2			Proposed	118	Kiunga-Aiambak		TA	Proposed
1824	Cape Bogan			Proposed					
1826	Wide Bay			Proposed					
1830	Taraiwara Ext			Proposed					
1838	Suikol			Proposed					